Small-Signal Stability of Sequence-Decomposed Grid-Forming IBRs with DC-Link Voltage Dynamics

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***Abstract*—The proliferation of technical specifications for grid-forming (GFM) inverter-based resources (IBRs) is accelerating globally to facilitate their effective integration in power grids. Notably, the precise regulation of negative sequence currents is currently mandated for GFM IBRs, necessitating the incorporation of sequence-decomposed modules (SDMs) into the GFM control architecture. However, SDMs have an adverse impact on the dynamics of the GFM control loops, resulting in performance loss of system stability. Specifically, GFM IBRs with DC-link voltage control (DVC) are susceptible to losing their steady-state operating points due to the presence of SDMs. This paper investigates the effects of SDMs on the small-signal stability of GFM IBRs with DVC. A single-input single-output small-signal model considering interactions of the entire control loops is developed to unveil the unstable sub-synchronous oscillation (SSO) issues induced by SDMs. Subsequently, potential remedies for the SSO issues are explored. To mitigate instability risks, an active power reference feedforward control method is proposed to actively dampen the oscillations inherent in the dynamics of active power and DC-link voltage. Controller hardware-in-the-loop (CHIL) experiments are conducted to corroborate the theoretical findings and the efficacy of the proposed control method under different grid conditions.**

***Index Terms*—Grid-forming control, sequence-decomposed control, DC-link voltage-synchronized control, small-signal stability, negative sequence current**

I. Introduction

GRID-FORMING (GFM) control has emerged as an effective solution for integrating high-penetration inverter-based resources (IBRs) into future power-electronics-dominated power systems [1], [2]. To efficiently deploy GFM IBRs in the power grid, well-defined GFM functional specifications are essential.

Recent technical specifications released for GFM IBRs outline their core performance criteria [3]-[8]. Apart from several fundamental capabilities such as grid support, positive damping, and fault ride-through, one notable requirement is the provision of negative sequence current under both normal operation and disturbances. Specifically, GFM IBRs are anticipated to provide negative sequence currents to maintain voltage balance in normal grids and to support reliable protection operation in faulted networks [6]-[8]. Therefore, it is crucial for GFM IBRs to regulate the sequence components of their control variables. This necessitates the adoption of the sequence-decomposed control in GFM IBRs [9], [10].

Prior research on sequence-decomposed GFM IBRs has primarily focused on enhancing fault ride-through capabilities to improve their output voltage or current profiles during grid faults [11]-[13]. Recent work in [14] sheds light on the small-signal stability of sequence-decomposed GFM IBRs, a facet previously overlooked. However, these studies often assume a constant DC-link voltage, neglecting its associated dynamics. In some cases, the DC-link voltage of wind- or solar-based IBRs leaves uncontrolled by front converters [15]. To implement GFM control in such applications, it is imperative to simultaneously consider both DC-link voltage regulation and synchronization. This requirement facilitates the development of DC-link voltage-synchronization control (DVSC) [16].

Existing studies indicate that GFM IBRs with DVSC are prone to low-frequency oscillation instabilities [17], [18]. To address these issues, DC-link voltage control (DVC) is typically cascaded or paralleled with active power control (APC) [19], [20]. IBRs using such control configuration are designated as GFM IBRs with DVC. Whereas a comprehensive small-signal stability analysis for sequence-decomposed GFM IBRs with DVC has not been reported thus far. Note that these sequence-decomposed GFM IBRs with DVC differ from their sequence-coupled counterparts due to their unique control architecture. Unlike the latter, the former requires the configuration of a sequence-decomposed module (SDM) in the control system to separate and extract positive and negative sequence components from the sampled signals [21], [22]. The introduction of SMDs perturbs control variables within each control loop, thereby reshaping the feedforward and feedback loops of the control system [23], [24]. Consequently, the overall control dynamics of the sequence-decomposed GFM IBRs with DVC become intricate and warrant further investigation.

Motivated by this target, the dynamic behaviors of SDMs are closely inspected in this paper to elaborate their impacts on the small-signal stability of the sequence-decomposed GFM IBR with DVC. The potential sub-synchronous oscillation (SSO) instability issues arising from the interactions between SDMs and other control loops are disclosed for the first time. An active

The work of Y. Liang and Z. Zhang was supported by the European Union's Horizon 2020 research and innovation programme under the Marie Sklodowska-Curie under Grant 861398. This work was supported in part by the U.K. Research and Innovation under the U.K. Government's Horizon Europe Funding Guarantee under Grant EP/X032477/1, and in part by, which is associated with the European Union's Horizon 2020 research and innovation programme under the Marie Sklodowska-Curie under Grant 101073558. *(Corresponding author: Xiaowei Zhao.)*

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power reference feedforward control scheme is proposed to enhance the stability robustness of sequence-decomposed GFM IBRs with DVC. Given the importance of operating points in small-signal stability analysis [25], [26], a wide range of operating conditions are examined in this study. Main contributions of this study are summarized as follows:

- A single-input single-output (SISO) small-signal analysis method is developed for sequence-decomposed GFM IBRs with DVC, accounting for the dynamics of the entire control loop.
- Comparative analyses are performed to evaluate the stability of sequence-coupled GFM IBRs with DVC, sequence-decomposed GFM IBRs with DVC, and GFM IBRs incorporating the proposed feedforward control.
- An active power reference feedforward control is proposed, designed with adequate stability margins to effectively mitigate the inherent SSO issues in sequence-decomposed GFM IBRs with DVC.

The remainder of the paper proceeds as follows. In Section II, the overall architecture of the studied system is illustrated. The SISO small-signal model of the sequence-decomposed GFM IBR with DVC is developed step-by-step in Section III. Then, stability analyses are conducted with varying DVC gains, APC droop gains, and active power setpoints in both weak and stiff AC grids. Section IV details the mechanism and design guideline for the proposed feedforward control. Validation analysis is performed in Section V through the controller hardware-in-the-loop (CHIL) experiments. Lastly, Section VI encapsulates the main results of this paper.

II. System Description

Fig. 1 depicts the single-line diagram of the sequence-decomposed GFM IBR with DVC, and the system parameters are provided in TABLE I. As documented in IEEE 2800-2022 [27], the control of the DC-side source is the slowest of the inverter control loops, with an operation rate varying from 0.5 s to around 5 s. In contrast, the typical operation rate for DVC is within one to six cycles. Due to these distinct control timescales, the DC-side source control has negligible impact on DVC. In this context, a constant power source is assumed at the DC-link [17], [18], [20] with the injected power P_s, representing the DC-side source of renewable energy sources like photovoltaic arrays or wind turbine generators. The power filter comprises the inverter-side inductor L_f, resistor R_f, and capacitor C_f, where the terminal voltage $\boldsymbol{v}$, inductor current $\boldsymbol{i_L}$ and point of common coupling (PCC) voltage $\boldsymbol{u}$ can be observed. Via the grid inductance L_g, the grid current $\boldsymbol{i_g}$ flows into the voltage source $\boldsymbol{v_g}$, representing the AC grid [28].

For the sequence-decomposed GFM IBR, DVC cascaded with APC is used as an illustrative example [19], [20], excluding instability issues inherent to DVSC [17], [18]. To alleviate impacts of the steady-state operation point of the DC-link voltage on the closed-loop dynamics, the square values of the DC-link voltage reference V_{dcref} and feedback DC-link voltage V_{dc} are selected to derive the active power reference P_{ref} [20], [29]. Within this framework, the control law of DVC is given by

$$P_{ref} = G_{DVC}\left(\left(V_{dc}\right)^2 - \left(V_{dcref}\right)^2\right) \tag{1}$$

where G_{DVC}= (k_{pdc}+k_{idc}/s), k_{pdc} and k_{idc} denote the proportional and integral gains of the DVC controller, respectively, and s is the Laplace operator.

The typical *P-f* and *Q-V* droop loops are employed for the APC and reactive power control (RPC) correspondingly, with D_p and D_q representing the respective droop gains.

To regulate the sequence components of voltage and current signals, the sequence-decomposed vector voltage control (VVC) and vector current control (VCC) are adopted with a typical synchronous *dq* frame. The separation of positive and negative sequence variables is accomplished using dual second-order generalized integrator (DSOGI) or notch filter (NF) based SDMs. These SDMs are chosen for their ability to accurately represent the dynamics of other SDM types [14], [21], [22]. This study excludes consideration of time-delay effects due to their negligible impact within the low-frequency range.

III. Small-Signal Stability Analysis of the Sequence-Decomposed GFM IBR with DVC

Based on Fig. 1, the synchronization stability of sequence-decomposed GFM IBR with DVC is determined by the DC-link voltage dynamics. Thus, the SISO mathematical relation from V_{dcref} to V_{dc} is derived for small-signal stability analysis.

A. Physical Circuit Dynamics

1) AC Circuit: Assuming a small disturbance is applied to the interconnection system, the linearization analysis can thus be conducted. The AC system *dq*-frame dynamics consist of AC grid dynamics and LC filter dynamics, and can be calculated via Fig. 1 as

$$\tilde{\boldsymbol{U}}^s_{dq} = \tilde{\boldsymbol{V}}^s_{gdq} + \boldsymbol{Z}_g\tilde{\boldsymbol{I}}^s_{gdq} = \boldsymbol{Z}_g\tilde{\boldsymbol{I}}^s_{gdq} \tag{2}$$

$$\tilde{\boldsymbol{I}}^s_{Ldq} = \tilde{\boldsymbol{I}}^s_{gdq} + \boldsymbol{Y}_c\tilde{\boldsymbol{U}}^s_{dq} = \boldsymbol{G}_c\tilde{\boldsymbol{I}}^s_{gdq} \tag{3}$$

$$\tilde{\boldsymbol{V}}^s_{dq} = \tilde{\boldsymbol{U}}^s_{dq} + \boldsymbol{Z}_f\tilde{\boldsymbol{I}}^s_{Ldq} = \left(\boldsymbol{Z}_g + \boldsymbol{Z}_f\boldsymbol{G}_c\right)\tilde{\boldsymbol{I}}^s_{gdq} \tag{4}$$

In (2)–(4), the small-signal variables are denoted by the symbol ~, variables with superscript s indicate their values in the system *dq*-frame. The bold letters with ~ represent the column vectors, i.e., $\tilde{\boldsymbol{U}}^s_{dq}$= [$\tilde{U}^{ps}_d$, $\tilde{U}^{ps}_q$, $\tilde{U}^{ns}_d$, $\tilde{U}^{ns}_q$]^T, $\tilde{\boldsymbol{V}}^s_{gdq}$= [$\tilde{V}^{ps}_{gd}$, $\tilde{V}^{ps}_{gq}$, $\tilde{V}^{ns}_{gd}$, $\tilde{V}^{ns}_{gq}$]^T, $\tilde{\boldsymbol{I}}^s_{gdq}$= [$\tilde{I}^{ps}_{gd}$, $\tilde{I}^{ps}_{gq}$, $\tilde{I}^{ns}_{gd}$, $\tilde{I}^{ns}_{gq}$]^T, $\tilde{\boldsymbol{I}}^s_{Ldq}$= [$\tilde{I}^{ps}_{Ld}$, $\tilde{I}^{ps}_{Lq}$, $\tilde{I}^{ns}_{Ld}$, $\tilde{I}^{ns}_{Lq}$]^T, $\tilde{\boldsymbol{V}}^s_{dq}$= [$\tilde{V}^{ps}_d$, $\tilde{V}^{ps}_q$, $\tilde{V}^{ns}_d$, $\tilde{V}^{ns}_q$]^T, where the positive and negative sequence values are described by superscripts p and n, respectively, and superscript T represents matrix transpose. Additionally, $\boldsymbol{Z}_g$, $\boldsymbol{Y}_c$, and $\boldsymbol{Z}_f$ are the 4×4 grid impedance matrix (L_g), 4×4 filter capacitor matrix (C_f), and 4×4 filter inductance and resistance matrix (L_f and R_f), respectively.

2) DC Circuit: The DC-link instantaneous power balance can be written in the small-signal functions as

$$\tilde{P}_{dc} = \tilde{P}_s - sC_d\left(\tilde{V}_{dc}\right)^2/2 = -sC_d\left(\tilde{V}_{dc}\right)^2/2 \tag{5}$$

To derive the small-signal dynamics of the DC-side output active power P_{dc}, linearization is essential. By considering the terminal voltage $\tilde{\boldsymbol{V}}^s_{dq}$ and inductor current $\tilde{\boldsymbol{I}}^s_{Ldq}$ in the system *dq*-

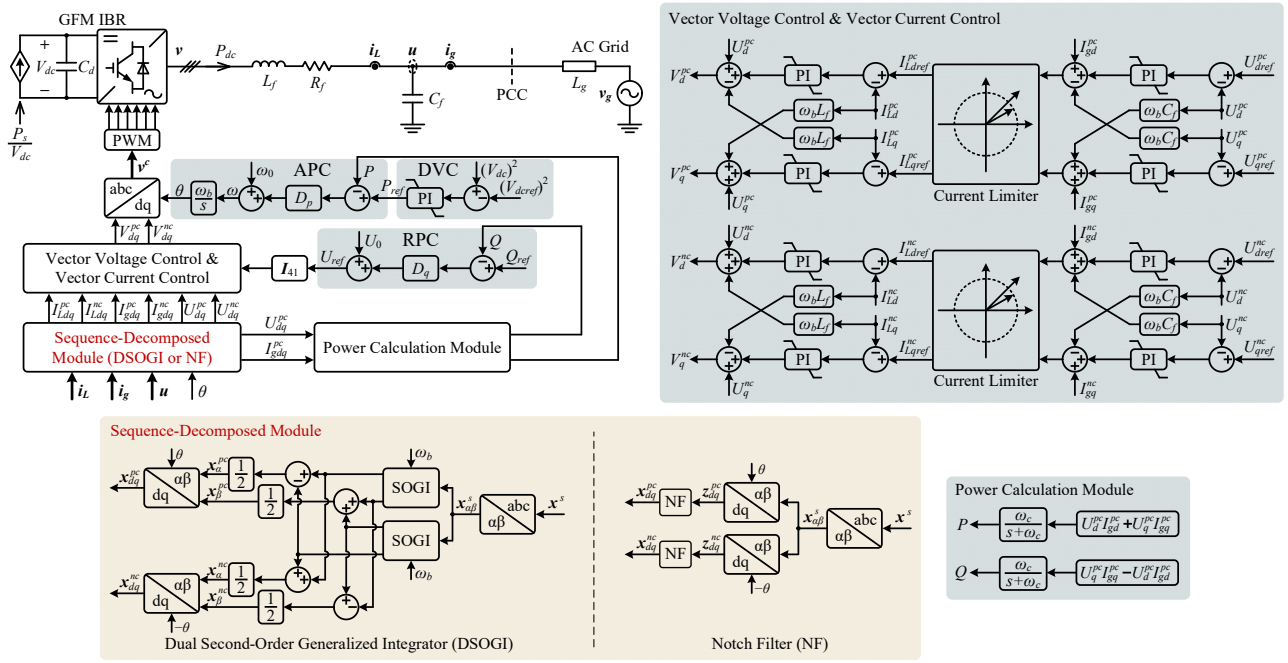


Fig. 1. Single-line diagram of the sequence-decomposed GFM IBR with DVC ($\boldsymbol{x}\in\{\boldsymbol{i}_L, \boldsymbol{i}_g, \boldsymbol{u}\}$).

TABLE I
MAIN SYSTEM PARAMETERS

Symbol	Description	Value
	GFM IBR	
S_b	Base power	15 kVA
V_b	Base voltage	310.27 V
ω_b	Base angular frequency	100π rad/s
L_f	LC filter inductor	6 mH
R_f	LC filter resistor	0.25 Ω
C_f	LC filter capacitor	75 μF
C_d	DC-link capacitor	2 mF
	Outer Control Loop	
ω_0	Nominal angular frequency	1 p.u.
U_0	Magnitude of PCC voltage	1 p.u.
P_s	DC-link injected power	0.7 p.u.
V_{dcref}	DC-link voltage reference	2.578 p.u.
Q_{ref}	Reactive power reference	0
D_p	APC droop gain	0.04 p.u.
D_q	RPC droop gain	0.1 p.u.
	Inner Control Loop	
k_{pv}/k_{iv}	VVC proportional/integral gain	0.324/2.078 p.u.
k_{pi}/k_{ii}	VCC proportional/integral gain	1.958/0.819 p.u.

frame, P_{dc} can be approximated as

$$P_{dc}=1.5\left(\boldsymbol{V}_d^{ps}\boldsymbol{I}_{Ld}^{ps}+\boldsymbol{V}_q^{ps}\boldsymbol{I}_{Lq}^{ps}\right) \tag{6}$$

By rewriting P_{dc} using its steady-state value P_{dc0} and small-signal perturbation $\tilde{P}_{dc}$, (6) becomes:

$$P_{dc}=P_{dc0}+\tilde{P}_{dc}=1.5\begin{bmatrix}\left(\boldsymbol{V}_{d0}^{ps}+\tilde{\boldsymbol{V}}_d^{ps}\right)\left(\boldsymbol{I}_{Ld0}^{ps}+\tilde{\boldsymbol{I}}_{Ld}^{ps}\right)+\\ \left(\boldsymbol{V}_{q0}^{ps}+\tilde{\boldsymbol{V}}_q^{ps}\right)\left(\boldsymbol{I}_{Lq0}^{ps}+\tilde{\boldsymbol{I}}_{Lq}^{ps}\right)\end{bmatrix} \tag{7}$$

where subscript 0 denotes the steady-state value.

Substituting the steady-state relation $P_{dc0}=V_{d0}^{ps}I_{Ld0}^{ps}+V_{q0}^{ps}I_{Lq0}^{ps}$ into (6), we obtain:

$$\tilde{P}_{dc}=1.5\begin{bmatrix}\left(\tilde{\boldsymbol{V}}_d^{ps}\boldsymbol{I}_{Ld0}^{ps}+\boldsymbol{V}_{d0}^{ps}\tilde{\boldsymbol{I}}_{Ld}^{ps}+\tilde{\boldsymbol{V}}_d^{ps}\tilde{\boldsymbol{I}}_{Ld}^{ps}\right)+\\ \left(\tilde{\boldsymbol{V}}_q^{ps}\boldsymbol{I}_{Lq0}^{ps}+\boldsymbol{V}_q^{ps}\tilde{\boldsymbol{I}}_{Lq}^{ps}+\tilde{\boldsymbol{V}}_q^{ps}\tilde{\boldsymbol{I}}_{Lq}^{ps}\right)\end{bmatrix} \tag{8}$$

The second-order small-signal terms $\tilde{V}_d^{ps}\tilde{I}_{Ld}^{ps}$ and $\tilde{V}_q^{ps}\tilde{I}_{Lq}^{ps}$ can be assumed negligible due to their minimal impact. As a result, the nonlinearity in (6) is eliminated, and $\tilde{P}_{dc}$ is shown as

$$\begin{aligned}\tilde{P}_{dc}&=1.5\left(\boldsymbol{G}_{Pd}^{u}\boldsymbol{I}_{24}\tilde{\boldsymbol{V}}_{dq}^{s}+\boldsymbol{G}_{Pd}^{i}\boldsymbol{I}_{24}\tilde{\boldsymbol{I}}_{Ldq}^{s}\right)=\boldsymbol{G}_{I_g-P_{dc}}\tilde{\boldsymbol{I}}_{gdq}^{s}\\ \boldsymbol{G}_{I_g-P_{dc}}&=1.5\left[\boldsymbol{G}_{Pd}^{u}\boldsymbol{I}_{24}\left(\boldsymbol{Z}_g+\boldsymbol{Z}_f\boldsymbol{G}_c\right)+\boldsymbol{G}_{Pd}^{i}\boldsymbol{I}_{24}\boldsymbol{G}_c\right]\end{aligned} \tag{9}$$

where $\boldsymbol{G}_{Pd}^{u}=[I_{Ld0}^{ps}, I_{Lq0}^{ps}]$, $\boldsymbol{G}_{Pd}^{i}=[V_{d0}^{ps}, V_{q0}^{ps}]$, and matrix $\boldsymbol{I}_{24}$ is

$$\boldsymbol{I}_{24}=\begin{bmatrix}1&0&0&0\\0&1&0&0\end{bmatrix} \tag{10}$$

B. Sequence-Decomposed Module Dynamics

Before considering the control loop dynamics, the sequence component $\boldsymbol{x}_{dq}^{c}$ needs to be generated by its three-phase variable $\boldsymbol{x}$ via the SDM, as presented in Fig. 1. In this case, positive and negative sequence control signals can be obtained in the controller *dq*-frame, denoted by superscript *c*.

1) DSOGI-Based SDM: With the control architecture of the DSOGI module shown in Fig. 1, mathematical relations between $\boldsymbol{x}_{dq}^{c}=[x_d^{pc}, x_q^{pc}, x_d^{nc}, x_q^{nc}]^T$ in the controller *dq*-frame and $\boldsymbol{x}_{dq}^{s}=[x_d^{ps}, x_q^{ps}, x_d^{ns}, x_q^{ns}]^T$ in the system *dq*-frame can be established in the time-domain as

$$\boldsymbol{x}_{dq}^{c}=\boldsymbol{T}(\theta)\underbrace{\left\{\mathcal{L}^{-1}[\boldsymbol{K}]*\left[\boldsymbol{C}\boldsymbol{T}^{-1}(\theta_s)\boldsymbol{x}_{dq}^{s}\right]\right\}}_{\boldsymbol{m}_{dq}^{s}} \tag{11}$$

where θ represents the phase angle generated by the GFM control system, $\theta_s=\omega_b t$ denotes that created by the AC grid, $\theta=\theta_s$ in the normal steady state, and

$$\boldsymbol{T}(\theta)=\left[\begin{array}{cc:cc}\cos(\theta)&\sin(\theta)&0&0\\-\sin(\theta)&\cos(\theta)&0&0\\\hdashline 0&0&\cos(\theta)&-\sin(\theta)\\0&0&\sin(\theta)&\cos(\theta)\end{array}\right] \tag{12}$$

$$\boldsymbol{K}=\frac{1}{2}\frac{k\omega_b}{s^2+k\omega_b s+\omega_b{}^2}\begin{bmatrix} s & \omega_b & s & -\omega_b \\ -\omega_b & s & \omega_b & s \end{bmatrix}^T \tag{13}$$

$$\boldsymbol{C}=\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \tag{14}$$

where k is the constant parameter used in the DSOGI module, with a typical value of $\sqrt{2}$ [30].

The rotation matrix $\boldsymbol{T}(\theta)$ introduces nonlinearity into the relation in (11). By decomposing the variables in (11) into their steady-state values and small-signal perturbations, we have:

$$\boldsymbol{x}^c_{dq0}+\tilde{\boldsymbol{x}}^c_{dq}=\boldsymbol{T}\left(\theta_s+\tilde{\theta}\right)\left(\boldsymbol{m}^s_{dq0}+\tilde{\boldsymbol{m}}^s_{dq}\right) \tag{15}$$

The rotation matrix in (15) can be linearized using

$$\begin{cases}\cos\left(\theta_s+\tilde{\theta}\right)\approx\cos\left(\theta_s\right)-\sin\left(\theta_s\right)\tilde{\theta}\\ \sin\left(\theta_s+\tilde{\theta}\right)\approx\sin\left(\theta_s\right)+\cos\left(\theta_s\right)\tilde{\theta}\end{cases} \tag{16}$$

By substituting (16) and the steady-state equation $\boldsymbol{x}^c_{dq0}=\boldsymbol{T}(\theta_s)\boldsymbol{m}^s_{dq0}$ into (15), and ignoring the resultant second-order small-signal terms, the small-signal expression is obtained as

$$\tilde{\boldsymbol{x}}^c_{dq}=\boldsymbol{T}\left(\theta_s\right)\tilde{\boldsymbol{m}}^s_{dq}+\boldsymbol{G}^\theta_X\tilde{\theta} \tag{17}$$

where $\boldsymbol{G}^\theta_X=[X^{pc}_{q0}, -X^{pc}_{d0}, -X^{nc}_{q0}, X^{nc}_{d0}]^T$, and $X^{pc}_{d0}, X^{pc}_{q0}, X^{nc}_{d0}, X^{nc}_{q0}$ are the steady-state values of $\boldsymbol{x}^c_{dq}$.

Assuming $\mathcal{L}\{f\}=F$, the modulation theory in [31] leads to the following equations:

$$\begin{cases}\mathcal{L}\left[f\cos\left(\theta_s\right)\right]=\dfrac{1}{2}\left(F\left(s-j\omega_b\right)+F\left(s+j\omega_b\right)\right)\\ \mathcal{L}\left[f\sin\left(\theta_s\right)\right]=\dfrac{1}{2j}\left(F\left(s-j\omega_b\right)-F\left(s+j\omega_b\right)\right)\end{cases} \tag{18}$$

By applying Laplace transform to both sides of (17) and using (18), the small-signal function of the DSOGI-based SDM is derived as

$$\tilde{\boldsymbol{X}}^c_{dq}=\boldsymbol{H}^{DSOGI}\tilde{\boldsymbol{X}}^s_{dq}+\tilde{\boldsymbol{X}}^{DSOGI}_{cp}+\boldsymbol{G}^\theta_X\tilde{\theta} \tag{19}$$

The term $\tilde{\boldsymbol{X}}^{DSOGI}_{cp}$ in (19) represents the frequency coupling dynamics caused by the rotation matrix, typically inducing a double frequency component in both the positive and negative sequences of the dq-frame. However, this frequency coupling effect is largely filtered out by the transfer functions formed by the DSOGI module and has negligible impact on the frequency range studied in this paper. Consequently, (19) is simplified as

$$\tilde{\boldsymbol{X}}^c_{dq}=\boldsymbol{H}^{DSOGI}\tilde{\boldsymbol{X}}^s_{dq}+\boldsymbol{G}^\theta_X\tilde{\theta}$$

$$\boldsymbol{H}^{DSOGI}=\frac{k\omega_b}{4H_d}\left[\begin{array}{cc:cc} H_1 & H_2 & 0 & 0 \\ -H_2 & H_1 & 0 & 0 \\ \hdashline 0 & 0 & H_1 & -H_2 \\ 0 & 0 & H_2 & H_1 \end{array}\right] \tag{20}$$

$$H_1=2s^3+2k\omega_b s^2+8\omega_b^2 s+4k\omega_b^3,\ H_2=-2k\omega_b^2 s$$

$$H_d=s^4+2k\omega_b s^3+\left(k^2+4\right)\omega_b^2 s^2+4k\omega_b^3 s+k^2\omega_b^4$$

2) NF-Based SDM: The mathematical expression of the NF-based SDM can be derived via Fig. 1 as

$$\boldsymbol{x}^c_{dq}=\mathcal{L}^{-1}\left[N\right]*\underbrace{\left[\boldsymbol{T}\left(\theta\right)\boldsymbol{C}^T\boldsymbol{C}\boldsymbol{T}^{-1}\left(\theta_s\right)\boldsymbol{x}^s_{dq}\right]}_{\boldsymbol{z}^c_{dq}} \tag{21}$$

N represents the transfer function of the NF, which is

$$N=\frac{s^2+\omega_{NF}{}^2}{s^2+2k_{NF}\omega_{NF}s+\omega_{NF}{}^2} \tag{22}$$

where the constant k_{NF} is normally selected as $\sqrt{2}/2$, and the center frequency ω_{NF} is 200π rad/s [21], [22].

Representing $\boldsymbol{z}^c_{dq}$ with its steady-state values and small-signal perturbations as

$$\boldsymbol{z}^c_{dq0}+\tilde{\boldsymbol{z}}^c_{dq}=\boldsymbol{T}\left(\theta_s+\tilde{\theta}\right)\boldsymbol{C}^T\boldsymbol{C}\boldsymbol{T}^{-1}\left(\theta_s\right)\left(\boldsymbol{x}^s_{dq0}+\tilde{\boldsymbol{x}}^s_{dq}\right) \tag{23}$$

By considering the steady-state relation $\boldsymbol{z}^c_{dq0}=\boldsymbol{T}(\theta_s)\boldsymbol{C}^T\boldsymbol{C}\boldsymbol{T}^{-1}(\theta_s)\boldsymbol{x}^s_{dq0}$ and following the same mathematical manipulations from (15) to (20), (23) can also be manipulated to obtain the small-signal expression of the NF-based SDM as

$$\tilde{\boldsymbol{X}}^c_{dq}=\boldsymbol{H}^{NF}\tilde{\boldsymbol{X}}^s_{dq}+\tilde{\boldsymbol{X}}^{NF}_{cp}+\boldsymbol{G}^\theta_X\tilde{\theta} \tag{24}$$

Note that $\tilde{\boldsymbol{X}}^{NF}_{cp}$ in (24) shares the same frequency coupling term with $\tilde{\boldsymbol{X}}^{DSOGI}_{cp}$, but $\tilde{\boldsymbol{X}}^{NF}_{cp}$ is multiplied by N rather than the transfer function from the DSOGI module. Thus, this frequency coupling effect is also neglected, as the NF shows significant attenuation for the double frequency components, resulting in:

$$\tilde{\boldsymbol{X}}^c_{dq}=\boldsymbol{H}^{NF}\tilde{\boldsymbol{X}}^s_{dq}+N\boldsymbol{G}^\theta_X\tilde{\theta}$$

$$\boldsymbol{H}^{NF}=\left[\begin{array}{cc:cc} N & 0 & 0 & 0 \\ 0 & N & 0 & 0 \\ \hdashline 0 & 0 & N & 0 \\ 0 & 0 & 0 & N \end{array}\right] \tag{25}$$

By using the matrix $\boldsymbol{H}$, a generic small-signal model of the SDM, free from frequency coupling dynamics, is yielded as

$$\tilde{\boldsymbol{X}}^c_{dq}=\boldsymbol{H}\tilde{\boldsymbol{X}}^s_{dq}+\boldsymbol{G}^\theta_X\tilde{\theta} \tag{26}$$

where $\boldsymbol{H}=\boldsymbol{H}^{DSOGI}$ or $\boldsymbol{H}^{NF}$, and $\boldsymbol{G}^\theta_X=[X^{pc}_{q0}, -X^{pc}_{d0}, -X^{nc}_{q0}, X^{nc}_{d0}]^T$ or $N[X^{pc}_{q0}, -X^{pc}_{d0}, -X^{nc}_{q0}, X^{nc}_{d0}]^T$, depending on the opted SDM type.

C. Overall Control Loop Dynamics

Based on the preparatory derivations of the physical circuit dynamics and SDM dynamics, the control loop dynamics can be independently deduced from Fig. 1, and expressed in the system dq-frame.

1) Power Calculation Module: The power calculation module determines the active power P and reactive power Q for the outer control loops using the PCC voltage $\tilde{\boldsymbol{U}}^c_{dq}$ and grid current $\tilde{\boldsymbol{I}}^c_{gdq}$ in the controller dq-frame. Following similar linearization procedures applied to P_{dc} from (6) to (9), the small-signal representations for P and Q are

$$\tilde{P}=1.5F_p\left(\boldsymbol{G}^u_P\boldsymbol{I}_{24}\tilde{\boldsymbol{U}}^c_{dq}+\boldsymbol{G}^i_P\boldsymbol{I}_{24}\tilde{\boldsymbol{I}}^c_{gdq}\right) \tag{27}$$

$$\tilde{Q}=1.5F_p\left(\boldsymbol{G}^u_Q\boldsymbol{I}_{24}\tilde{\boldsymbol{U}}^c_{dq}+\boldsymbol{G}^i_Q\boldsymbol{I}_{24}\tilde{\boldsymbol{I}}^c_{gdq}\right) \tag{28}$$

where $\boldsymbol{G}^u_P=[I^{pc}_{gd0}, I^{pc}_{gq0}]$, $\boldsymbol{G}^i_P=[U^{pc}_{d0}, U^{pc}_{q0}]$, $\boldsymbol{G}^u_Q=[-I^{pc}_{gq0}, I^{pc}_{gd0}]$, $\boldsymbol{G}^i_Q=[U^{pc}_{q0}, -U^{pc}_{d0}]$, and F_p represents the low-pass filter (LPF) with cut-off frequency ω_c (10 Hz in this study).

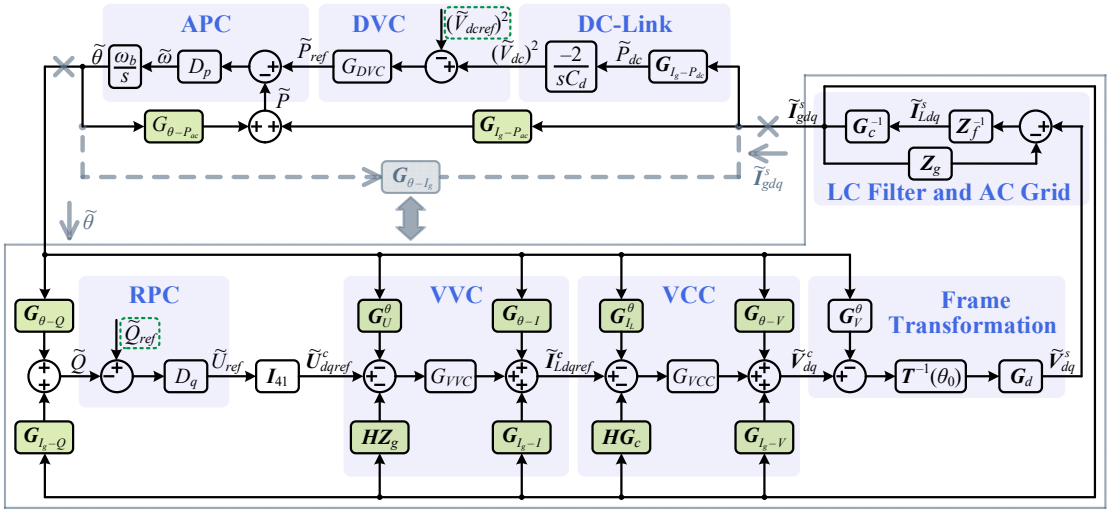


Fig. 2. Block diagram of the small-signal model of the sequence-decomposed GFM IBR with DVC (The impact of SDM is highlighted with green backgrounds).

By substituting (2) and (26) into (27) and (28), $\tilde{P}$ and $\tilde{Q}$ can be derived as

$$\tilde{P} = \boldsymbol{G}_{I_g-P_{ac}}\tilde{\boldsymbol{I}}^s_{gdq} + G_{\theta-P_{ac}}\tilde{\theta}$$

$$\begin{cases} \boldsymbol{G}_{I_g-P_{ac}} = 1.5F_p\left(\boldsymbol{G}^u_P\boldsymbol{I}_{24}\boldsymbol{H}\boldsymbol{Z}_g + \boldsymbol{G}^i_P\boldsymbol{I}_{24}\boldsymbol{H}\right) \\ G_{\theta-P_{ac}} = 1.5F_p\left(\boldsymbol{G}^u_P\boldsymbol{I}_{24}\boldsymbol{G}^\theta_U + \boldsymbol{G}^i_P\boldsymbol{I}_{24}\boldsymbol{G}^\theta_{I_g}\right) \end{cases} \tag{29}$$

$$\tilde{Q} = \boldsymbol{G}_{I_g-Q}\tilde{\boldsymbol{I}}^s_{gdq} + G_{\theta-Q}\tilde{\theta}$$

$$\begin{cases} \boldsymbol{G}_{I_g-Q} = 1.5F_p\left(\boldsymbol{G}^u_Q\boldsymbol{I}_{24}\boldsymbol{H}\boldsymbol{Z}_g + \boldsymbol{G}^i_Q\boldsymbol{I}_{24}\boldsymbol{H}\right) \\ G_{\theta-Q} = 1.5F_p\left(\boldsymbol{G}^u_Q\boldsymbol{I}_{24}\boldsymbol{G}^\theta_U + \boldsymbol{G}^i_Q\boldsymbol{I}_{24}\boldsymbol{G}^\theta_{I_g}\right) \end{cases} \tag{30}$$

2) Outer Control Loops: The outer control loops encompass the DVC loop, APC loop and RPC loop, each characterized by their respective small-signal expressions, as follows:

$$\tilde{P}_{ref} = G_{DVC}\left(\left(\tilde{V}_{dc}\right)^2 - \left(\tilde{V}_{dcref}\right)^2\right) \tag{31}$$

$$\tilde{\theta} = \omega_b D_p\left(\tilde{P}_{ref} - \tilde{P}\right)/s \tag{32}$$

$$\tilde{U}_{ref} = D_q\left(\tilde{Q}_{ref} - \tilde{Q}\right) \tag{33}$$

3) Inner Control Loops: The inner voltage and current control loops are symmetric, and can be expressed as

$$\tilde{\boldsymbol{I}}^c_{Ldqref} = G_{VVC}\left(\tilde{\boldsymbol{U}}^c_{dqref} - \tilde{\boldsymbol{U}}^c_{dq}\right) + \boldsymbol{J}\omega_b C_f\tilde{\boldsymbol{U}}^c_{dq} + \tilde{\boldsymbol{I}}^c_{gdq} \tag{34}$$

$$\tilde{\boldsymbol{V}}^c_{dq} = G_{VCC}\left(\tilde{\boldsymbol{I}}^c_{Ldqref} - \tilde{\boldsymbol{I}}^c_{Ldq}\right) + \boldsymbol{J}\omega_b L_f\tilde{\boldsymbol{I}}^c_{Ldq} + \tilde{\boldsymbol{U}}^c_{dq} \tag{35}$$

where $\tilde{\boldsymbol{U}}^c_{dqref} = \boldsymbol{I}_{41}\tilde{U}_{ref}$, and $\boldsymbol{I}_{41} = [1, 0, 0, 0]^T$. $G_{VVC} = (k_{pv} + k_{iv}/s)$, $G_{VCC} = (k_{pi} + k_{ii}/s)$, and $\boldsymbol{J}$ is the 4×4 constant matrix.

By applying (2), (3) and (26) to (34) and (35), $\tilde{\boldsymbol{I}}^c_{Ldqref}$ and $\tilde{\boldsymbol{V}}^c_{dq}$ are rendered to

$$\tilde{\boldsymbol{I}}^c_{Ldqref} = G_{VVC}\left[\tilde{\boldsymbol{U}}^c_{dqref} - \left(\boldsymbol{H}\boldsymbol{Z}_g\tilde{\boldsymbol{I}}^s_{gdq} + \boldsymbol{G}^\theta_U\tilde{\theta}\right)\right] + \boldsymbol{G}_{I_g-I}\tilde{\boldsymbol{I}}^s_{gdq} + \boldsymbol{G}_{\theta-I}\tilde{\theta} \tag{36}$$

$$\begin{cases} \boldsymbol{G}_{I_g-I} = \boldsymbol{J}\omega_b C_f\boldsymbol{H}\boldsymbol{Z}_g + \boldsymbol{H} \\ \boldsymbol{G}_{\theta-I} = \boldsymbol{J}\omega_b C_f\boldsymbol{G}^\theta_U + \boldsymbol{G}^\theta_{I_g} \end{cases}$$

$$\tilde{\boldsymbol{V}}^c_{dq} = G_{VCC}\left[\tilde{\boldsymbol{I}}^c_{Ldqref} - \left(\boldsymbol{H}\boldsymbol{G}_c\tilde{\boldsymbol{I}}^s_{gdq} + \boldsymbol{G}^\theta_{I_L}\tilde{\theta}\right)\right] + \boldsymbol{G}_{I_g-V}\tilde{\boldsymbol{I}}^s_{gdq} + \boldsymbol{G}_{\theta-V}\tilde{\theta} \tag{37}$$

$$\begin{cases} \boldsymbol{G}_{I_g-V} = \boldsymbol{J}\omega_b L_f\boldsymbol{H}\boldsymbol{G}_c + \boldsymbol{H}\boldsymbol{Z}_g \\ \boldsymbol{G}_{\theta-V} = \boldsymbol{J}\omega_b L_f\boldsymbol{G}^\theta_{I_L} + \boldsymbol{G}^\theta_U \end{cases}$$

Considering the time delay matrix $\boldsymbol{G}_d$ and frame transform dynamics, the modulation voltage $\tilde{\boldsymbol{V}}^c_{dq}$ is further inferred as

$$\tilde{\boldsymbol{V}}^c_{dq} = \boldsymbol{T}\left(\theta_0\right)\boldsymbol{G}_d^{-1}\tilde{\boldsymbol{V}}^s_{dq} + \boldsymbol{G}^\theta_V\tilde{\theta} \tag{38}$$

where $\boldsymbol{G}_d \approx \boldsymbol{I}_{44}$, and $\theta_0 = 0$ in most cases.

D. Closed-Loop System Dynamics

1) Multi-Input Multi-Output System Dynamics: The complete small-signal dynamics of the sequence-decomposed GFM IBR with DVC can be recovered from the power circuit dynamics in (2)–(9) and the control dynamics in (26)–(38), as shown in Fig. 2. The closed-loop system in Fig. 2 represents a multi-output multi-input (MIMO) system with two reference inputs $(\tilde{V}_{dcref})^2$ and $\tilde{Q}_{ref}$, highlighted in green dotted boxes. The observation of Fig. 2 demonstrates that the SDM introduces the matrix $\boldsymbol{H}$ into the control loop interactions, including the APC, RPC, VVC, and VCC. Even though the DVC and DC-link loops are not directly influenced by $\boldsymbol{H}$, effects from $\boldsymbol{H}$ are transmitted to them via the *dq*-frame variables. By recalling (20) and (25), it can be intuitively seen that $\boldsymbol{H}$ covers the cross-coupled dynamics in the *dq*-frame, thus, the frequency response of the GFM IBR is reshaped by the SDM.

To facilitate the analysis, $(\tilde{V}_{dc})^2$ and $\tilde{Q}$ are selected as the MIMO system's outputs. Equations (9) and (30) show that the outputs $(\tilde{V}_{dc})^2$ and $\tilde{Q}$ are related to the grid current dynamics $\tilde{\boldsymbol{I}}^s_{gdq}$ and the phase angle dynamics $\tilde{\theta}$. By expressing $\tilde{\boldsymbol{I}}^s_{gdq}$ and $\tilde{\theta}$ in terms of the inputs $(\tilde{V}_{dcref})^2$ and $\tilde{Q}_{ref}$ or the outputs $(\tilde{V}_{dc})^2$ and $\tilde{Q}$, the closed-loop system in Fig. 2 is equivalently transformed into several individual transfer functions considering couplings between $(\tilde{V}_{dc})^2$ and $\tilde{Q}$. Specifically, (4), (33), (37), and (38) are combined to derive $\tilde{\boldsymbol{I}}^s_{gdq}$, and $\tilde{\theta}$ is deduced by substituting (29)

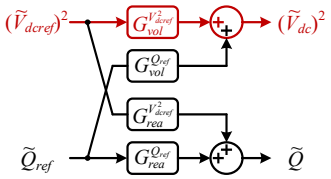


Fig. 3. MIMO model of the sequence-decomposed GFM IBR with DVC.

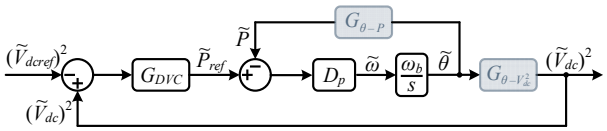


Fig. 4. SISO model of the sequence-decomposed GFM IBR with DVC.

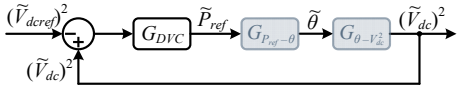


Fig. 5. Simplified SISO model of the sequence-decomposed GFM IBR with DVC.

and (31) into (32). With the equations of $\tilde{\boldsymbol{I}}_{gdq}^{s}$ and $\tilde{\theta}$, the DC-link voltage dynamics $(\tilde{V}_{dc})^2$ in (9) can be rearranged using $(\tilde{V}_{dcref})^2$, $\tilde{Q}_{ref}$, and $\tilde{Q}$, while the reactive power dynamics $\tilde{Q}$ in (30) can be represented using $(\tilde{V}_{dcref})^2$, $\tilde{Q}_{ref}$, and $(\tilde{V}_{dc})^2$. By substituting the derived $(\tilde{V}_{dc})^2$ and $\tilde{Q}$ into each other, interactions can be eliminated, and eventually, the closed-loop DC-link voltage and reactive power dynamics are shown in Fig. 3.

The derivation for Fig. 3 considers both multiple sequences (positive and negative) and multiple channels (d and q), as indicated by the variable definitions (e.g., $\tilde{\boldsymbol{I}}_{gdq}^{s}$). Fig. 3 demonstrates that the overall system stability can be analyzed using four individual closed-loop transfer functions: $G_{vol}^{V_{dcref}^2}$, $G_{vol}^{Q_{ref}}$, $G_{rea}^{V_{dcref}^2}$, and $G_{rea}^{Q_{ref}}$. Each transfer function can be considered as a SISO closed-loop system. By replacing these transfer functions with their respective elements in Section III, it can be found that they share the same denominator polynomials. Therefore, a single SISO stability analysis of any of $G_{vol}^{V_{dcref}^2}$, $G_{vol}^{Q_{ref}}$, $G_{rea}^{V_{dcref}^2}$, or $G_{rea}^{Q_{ref}}$ is sufficient to evaluate the entire system's stability.

2) Single-Input Single-Output System Dynamics: Since a single SISO closed-loop transfer function in Fig. 3 can feasibly analyze the MIMO system's small-signal stability, $\tilde{Q}_{ref}= 0$ can then be assumed. In this context, the DC-link voltage dynamics from $(\tilde{V}_{dcref})^2$ to $(\tilde{V}_{dc})^2$ are considered to develop the open-loop and closed-loop transfer functions of the SISO model for stability analysis and control design (highlighted in red in Fig. 3). Notably, many other approaches for SISO simplification exist, such as SISO models simplified under specific operating points [32], [33] or general MIMO models with SISO stability criteria [34], [35]. However, this paper does not focus on specific stability analysis methods, as these are well-established. Instead, it aims to explore instability issues arising from sequence-decomposed control in GFM IBRs with DVC, an area that is underexplored in the current literature.

By applying $\tilde{Q}_{ref}= 0$, the slate blue solid lines in Fig. 2 are first reorganized using a unified function $\boldsymbol{G}_{\theta-I_g}= \tilde{\boldsymbol{I}}_{gdq}^{s}/\tilde{\theta}$ in (39).

$$\boldsymbol{G}_{\theta-I_g}=\begin{Bmatrix}\boldsymbol{G}_{I_g-V}-\boldsymbol{T}(\theta_0)\boldsymbol{G}_d^{-1}\left(\boldsymbol{Z}_f\boldsymbol{G}_c+\boldsymbol{Z}_g\right)+\\ G_{VCC}\begin{bmatrix}G_{VVC}\left(\boldsymbol{I}_{41}D_q\boldsymbol{G}_{I_g-Q}-\boldsymbol{H}\boldsymbol{Z}_g\right)+\\ \boldsymbol{G}_{I_g-I}-\boldsymbol{H}\boldsymbol{G}_c\end{bmatrix}\end{Bmatrix}^{-1}\cdot \begin{Bmatrix}-\boldsymbol{G}_{\theta-V}+\boldsymbol{G}_V^{\theta}-\\ G_{VCC}\begin{bmatrix}G_{VVC}\left(\boldsymbol{I}_{41}D_q\boldsymbol{G}_{\theta-Q}-\boldsymbol{G}_U^{\theta}\right)+\\ \boldsymbol{G}_{\theta-I}-\boldsymbol{G}_{I_L}^{\theta}\end{bmatrix}\end{Bmatrix} \tag{39}$$

Then, $\tilde{P}$ and $(\tilde{V}_{dc})^2$ can be represented using the phase angle dynamics as

$$\left(\tilde{V}_{dc}\right)^2=G_{\theta-V_{dc}^2}\tilde{\theta}=-2\boldsymbol{G}_{I_g-P_{dc}}\boldsymbol{G}_{\theta-I_g}\tilde{\theta}/\left(sC_d\right) \tag{40}$$

$$\tilde{P}=G_{\theta-P}\tilde{\theta}=\left(G_{\theta-P_{ac}}+\boldsymbol{G}_{I_g-P_{ac}}\boldsymbol{G}_{\theta-I_g}\right)\tilde{\theta} \tag{41}$$

Based on (39)–(41), Fig. 2 is further simplified as Fig. 4.

Finally, by rearranging the loop from $\tilde{\theta}$ to $\tilde{P}$, a unit negative feedback model can be obtained in Fig. 5, where

$$G_{P_{ref}-\theta}=\tilde{\theta}/\tilde{P}_{ref}=\omega_b D_p/\left(s+\omega_b D_p G_{\theta-P}\right) \tag{42}$$

The open-loop and closed-loop transfer functions of the SISO model from $(\tilde{V}_{dcref})^2$ to $(\tilde{V}_{dc})^2$ are determined as

$$TF_{ol}=-G_{\theta-V_{dc}^2}G_{P_{ref}-\theta}G_{DVC},\quad TF_{cl}=TF_{ol}/\left(1+TF_{ol}\right) \tag{43}$$

The accuracy of the model in (43) is validated in Section V via CHIL experiments. Using the small-signal model in (43), the stability margin of the entire system can be determined directly from the open-loop transfer function. Meanwhile, the DC-link voltage dynamics of the sequence-decomposed GFM IBR with DVC can be accurately characterized by the closed-loop transfer function in (43), such as the step response of the DC-link voltage and the pole loci of the system. Therefore, system stability can be evaluated either by applying the Nyquist stability criterion to the open-loop transfer function or by analyzing the pole trajectories obtained from the closed-loop transfer function. In this paper, system pole trajectories are analyzed in Section III.E to identify the determinate unstable mode introduced by sequence-decomposed control in GFM IBRs with DVC. Subsequently, Nyquist plots are shown in Section IV to guide the parameter design of the proposed feedforward control by intuitively reflecting stability margins. This dual approach provides a systematic perspective on the stability analysis and control design of sequence-decomposed GFM IBRs with DVC.

Assuming $(\tilde{V}_{dcref})^2= 0$ and $\tilde{Q}_{ref}= 0$, the model in Fig. 2 can also be reformulated as either a MIMO output-impedance $\boldsymbol{Z}_{out}$ or a MIMO input-admittance model $\boldsymbol{Y}_{out}$ by replacing $\boldsymbol{Z}_g\tilde{\boldsymbol{I}}_{gdq}^{s}$ with $\tilde{\boldsymbol{U}}_{dq}^{s}$, as shown in (44). The generalized Nyquist criterion can then be applied to analyze system stability. Given that $\boldsymbol{H}^{DSOGI}$ and $\boldsymbol{H}^{NF}$ in (20) and (25) eliminate frequency coupling terms, the coupling effects are confined to the dq channels of each respective sequence. Consequently, $Y_{dd}^{pn}= Y_{dq}^{pn}= Y_{qd}^{pn}= Y_{qq}^{pn}= 0$ and $Y_{dd}^{np}= Y_{dq}^{np}= Y_{qd}^{np}= Y_{qq}^{np}= 0$. Additionally, the

asymmetric control architectures of the outer control loops (DVC, APC, and RPC) may also induce frequency coupling dynamics in (44) under certain conditions. By transforming the space vector representations in (44) into complex space vectors, the associated frequency coupling terms can be revealed. For a more detailed discussion on frequency couplings and harmonic analysis, please refer to [36]-[39].

$$\tilde{\boldsymbol{I}}_{gdq}^{s} = -\boldsymbol{Y}_{out}\tilde{\boldsymbol{U}}_{dq}^{s},\ \boldsymbol{Z}_{out} = \boldsymbol{Y}_{out}^{-1}$$

$$\boldsymbol{Y}_{out} = \left[\begin{array}{cc|cc} Y_{dd}^{pp} & Y_{dq}^{pp} & Y_{dd}^{pn} & Y_{dq}^{pn} \\ Y_{qd}^{pp} & Y_{qq}^{pp} & Y_{qd}^{pn} & Y_{qq}^{pn} \\ \hline Y_{dd}^{np} & Y_{dq}^{np} & Y_{dd}^{nn} & Y_{dq}^{nn} \\ Y_{qd}^{np} & Y_{qq}^{np} & Y_{qd}^{nn} & Y_{qq}^{nn} \end{array}\right] \tag{44}$$

E. Comparative Small-Signal Stability Analysis of the GFM IBR Without and with the SDM

In this subsection, the closed-loop transfer function of the small-signal model from (43) is used to calculate the system pole trajectories. The analysis considers GFM IBRs without SDMs, as well as those with the DSOGI- and NF-based SDMs, to provide a comprehensive assessment of system stability.

1): Equilibrium Point Computation: To conduct stability analysis using the proposed model in (43), it is necessary to first compute the equilibrium points in the *dq*-frame, including the terminal voltage $\boldsymbol{v}$, PCC voltage $\boldsymbol{u}$, grid current $\boldsymbol{i_g}$, and inductor current $\boldsymbol{i_L}$, as utilized in the model in Fig. 2.

During normal steady-state operation, the *dq*-axis PCC voltages are obtained as $\boldsymbol{U}_{dq0}^{s} = \boldsymbol{U}_{dq0}^{c} = [U_{d0}^{pc}, U_{q0}^{pc}, U_{d0}^{nc}, U_{q0}^{nc}]^T = [U_{ref}, 0, 0, 0]^T$, based on (34) and the control law of VVC. Consequently, the *dq*-axis grid currents $\boldsymbol{I}_{gdq0}^{s} = \boldsymbol{I}_{gdq0}^{c} = [I_{gd0}^{pc}, I_{gq0}^{pc}, I_{gd0}^{nc}, I_{gq0}^{nc}]^T$ can be expressed as

$$\begin{cases} I_{gd0}^{pc} = P_0 / U_{ref},\ I_{gq0}^{pc} = -Q_0 / U_{ref} \\ I_{gd0}^{nc} = 0,\ I_{gq0}^{nc} = 0 \end{cases} \tag{45}$$

Given that the sequence-decomposed GFM IBR with DVC operates at the fundamental frequency under normal operations, i.e., $\omega = \omega_0$, the steady-state active power is $P_0 = P_{ref} = P_s$ in accordance with the droop law of the APC in Fig. 1. Additionally, the steady-state reactive power can be derived from the droop law of the RPC as

$$Q_0 = Q_{ref} - \left(U_{ref} - U_0\right) / D_q \tag{46}$$

Assuming a power angle δ between the PCC voltage and the grid voltage, $\sin(\delta)$ and $\cos(\delta)$ are given by

$$\begin{cases} \sin\left(\delta\right) = P_0 X_g / \left(U_{ref} V_g\right) \\ \cos\left(\delta\right) = \sqrt{1 - \left(\sin\left(\delta\right)\right)^2} \end{cases} \tag{47}$$

where $X_g = \omega_0 L_g$.

In line with the Kirchhoff's voltage law, the *d*-axis positive sequence PCC voltage can be expressed in terms of the grid voltage as

$$U_{d0}^{pc} = U_{ref} = -X_g I_{gq0}^{pc} + V_g \cos\left(\delta\right) \tag{48}$$

By substituting (45)–(47) into (48), the equilibrium points for the PCC voltage are determined. Subsequently, the equilibrium points for the grid current can be calculated using (45). With these values, the steady-state inductor current and terminal voltage can be readily computed based on Kirchhoff's current and voltage laws over the LC power filter.

2) Stability Analysis Under Different Control Gains of the DVC and APC: As shown in Fig. 1, synchronization dynamics are closely related to the control dynamics of DVC and APC. Therefore, a range of proportional gains (k_{pdc}) for DVC and droop gains (D_p) for APC are considered for stability analysis under both weak ($L_g = 0.85$ p.u., SCR= 1.2) and strong ($L_g = 0.2$ p.u., SCR= 5) grid conditions. Based on the parameters in TABLE I and the equations provided in *1)*, equilibrium points are calculated for various grid conditions. The pole loci are then derived using TF_{cl} in (43), as shown in Figs. 6–9. A pair of sub-synchronous frequency conjugate poles, denoted as "λ_1" and "λ_2", dominate system stability.

Fig. 6 and Fig. 7 show the pole trajectories for varying values of k_{pdc} under weak and strong grid conditions, respectively. The proportional gain k_{pdc} of DVC varies from 0.02 p.u. to 0.15 p.u., corresponding to a typical control bandwidth of DVC within 10 Hz [18], [40]. Meanwhile, the integral gain of DVC is set as $k_{idc} = k_{pdc}\omega_{dc}/10$ to mitigate its phase lag effect [18], [41], with $\omega_{dc} = 10\pi$ representing the cut-off frequency. The available k_{pdc} values in Fig. 6(a), (b), and (c) range from 0.02 p.u. to 0.15 p.u., indicating no instability issues under weak grid conditions for all three cases. Additionally, a moderate k_{pdc} value around 0.07 p.u. can even slightly improve small-signal stability in a stiff grid for the sequence-decomposed GFM IBR, as seen in Fig. 6(b) and Fig. 6(c). Since DVC is cascaded with APC in this study, a negative feedback loop ($G_{\theta-P}$ in Fig. 4), is formed to dampen potential oscillations. Consequently, the GFM IBR without SDMs demonstrates robust stability even under strong grid conditions, with optional k_{pdc} values up to 0.15 p.u., as shown in Fig. 7(a). Whereas all SSO mode conjugate poles $\lambda_{1\text{-}2}$ of the DSOGI-based GFM IBR with DVC in Fig. 7(b) are located in the right-half plane (RHP), indicating a lack of stable operating points and high susceptibility to SSO issues in stiff grids. Similarly, Fig. 7(c) shows that the NF-based SDM also degrades system stability, limiting the maximum k_{pdc} value to only 0.1 p.u. As a result, the NF-based GFM IBR with DVC is also vulnerable to SSO risks under strong grid conditions.

Figs. 8 and 9 display the pole loci results for various droop gains (D_p) of the APC. The variation range of D_p is 0.02-0.05 p.u., as specified by IEEE Standard 2800-2022 [27], and k_{pdc} is set at 0.1 p.u. As illustrated in Fig. 8, a robust system stability can be easily achieved for all D_p values in a weak grid. The stability robustness across the three cases in Fig. 8(a), (b), and (c) remains comparable for the same D_p settings. Unlike the analysis results under weak grid conditions, an increase in D_p tends to exacerbate SSO issues in a stiff grid, as indicated in Fig. 9. Specifically, the DSOGI-based GFM IBR with DVC is particularly vulnerable to SSO instability, as evidenced by the absence of stable D_p values in Fig. 9(b). In comparison, the NF-based GFM IBR with DVC shows relatively better system stability with multiple stable D_p values seen in Fig. 9(c), though it still underperforms compared to the GFM IBR without SDMs in Fig. 9(a). This highlights the detrimental effects of SDMs on

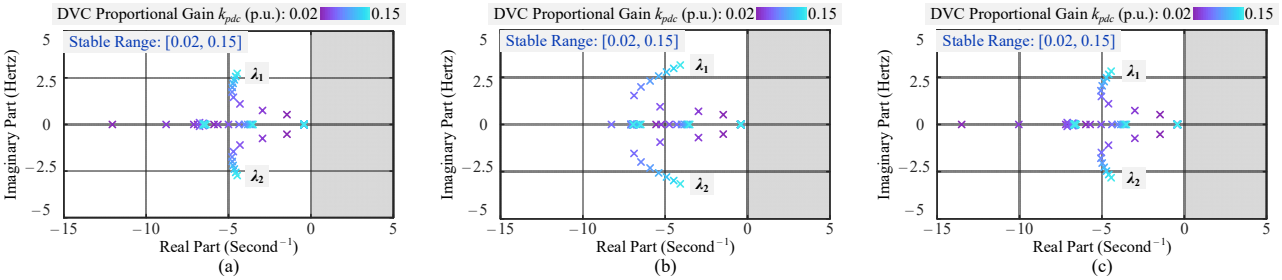


Fig. 6. Closed-loop poles of the SISO model under the weak grid condition with varying k_{pdc} (V_{dcref}= 2.578 p.u., P_{ref}= 0.7 p.u., Q_{ref}= 0, L_g= 0.85 p.u., D_p= 0.04 p.u.). (a) GFM IBR without SDMs. (b) GFM IBR with the DSOGI-based SDM. (c) GFM IBR with the NF-based SDM.

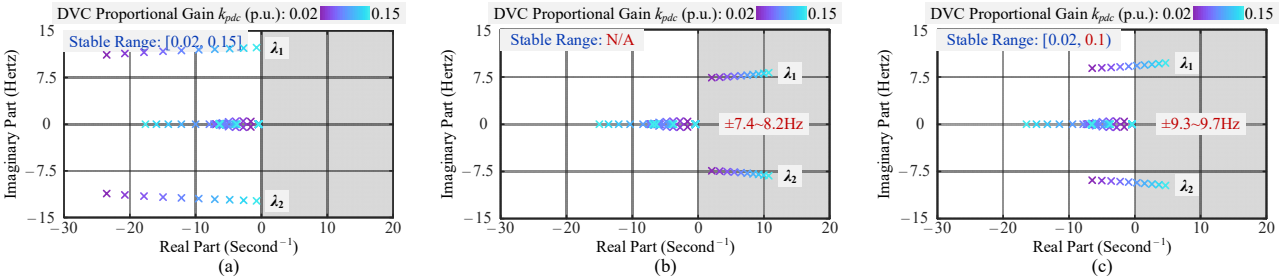


Fig. 7. Closed-loop poles of the SISO model under the strong grid condition with varying k_{pdc} (V_{dcref}= 2.578 p.u., P_{ref}= 0.7 p.u., Q_{ref}= 0, L_g= 0.2 p.u., D_p= 0.04 p.u.). (a) GFM IBR without SDMs. (b) GFM IBR with the DSOGI-based SDM. (c) GFM IBR with the NF-based SDM.

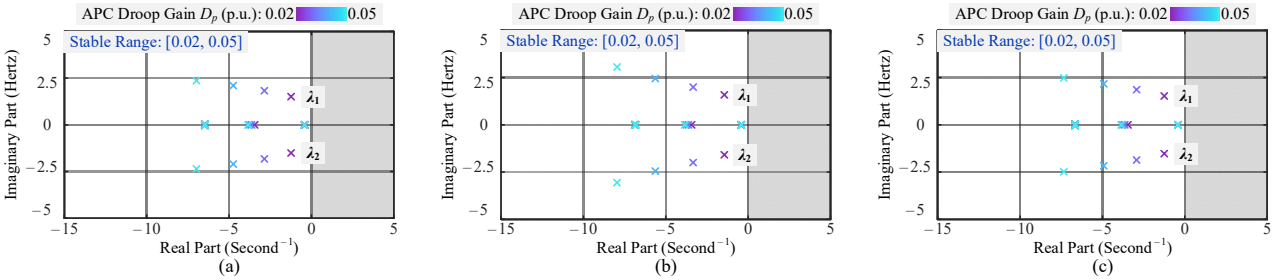


Fig. 8. Closed-loop poles of the SISO model under the weak grid condition with varying D_p (V_{dcref}= 2.578 p.u., P_{ref}= 0.7 p.u., Q_{ref}= 0, L_g= 0.85 p.u., k_{pdc}= 0.1 p.u.). (a) GFM IBR without SDMs. (b) GFM IBR with the DSOGI-based SDM. (c) GFM IBR with the NF-based SDM.

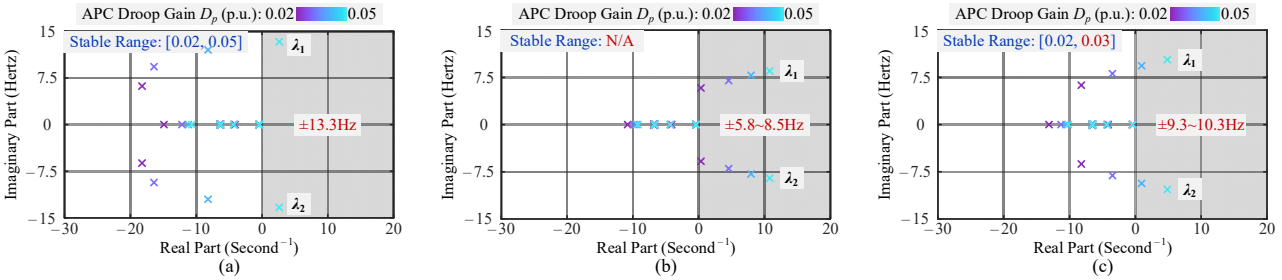


Fig. 9. Closed-loop poles of the SISO model under the strong grid condition with varying D_p (V_{dcref}= 2.578 p.u., P_{ref}= 0.7 p.u., Q_{ref}= 0, L_g= 0.2 p.u., k_{pdc}= 0.1 p.u.). (a) GFM IBR without SDMs. (b) GFM IBR with the DSOGI-based SDM. (c) GFM IBR with the NF-based SDM.

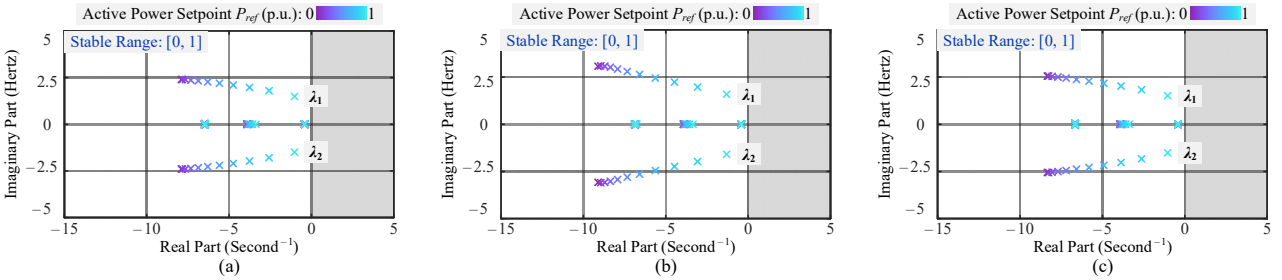


Fig. 10. Closed-loop poles of the SISO model under the weak grid condition with varying P_{ref} (V_{dcref}= 2.578 p.u., Q_{ref}= 0, L_g= 0.85 p.u., D_p= 0.04 p.u., k_{pdc}= 0.1 p.u.). (a) GFM IBR without SDMs. (b) GFM IBR with the DSOGI-based SDM. (c) GFM IBR with the NF-based SDM.

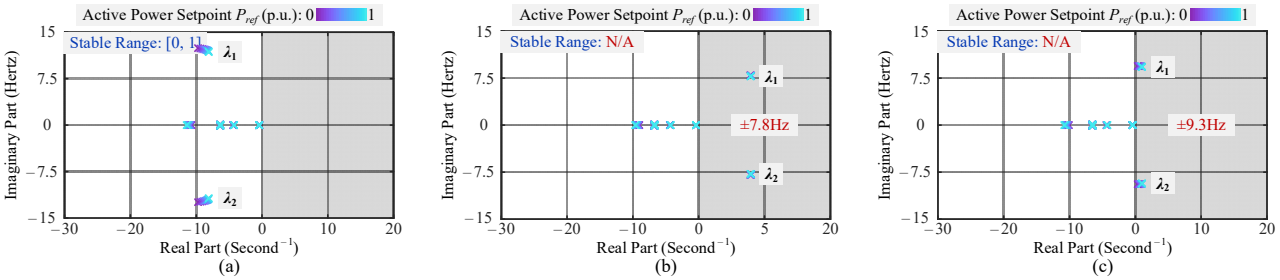


Fig. 11. Closed-loop poles of the SISO model under the strong grid condition with varying P_{ref} (V_{dcref}= 2.578 p.u., Q_{ref}= 0, L_g= 0.2 p.u., D_p= 0.04 p.u., k_{pdc}= 0.1 p.u.). (a) GFM IBR without SDMs. (b) GFM IBR with the DSOGI-based SDM. (c) GFM IBR with the NF-based SDM.

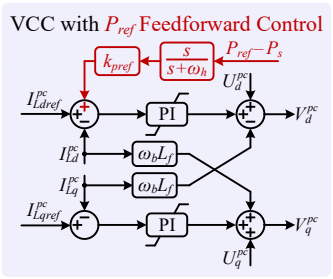


Fig. 12. Proposed P_{ref} feedforward control based on the active power reference derived from DVC.

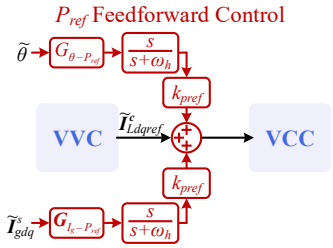


Fig. 13. Small-signal model of the proposed P_{ref} feedforward control.

system stability under strong grid conditions.

3) Stability Analysis Under Different Active Power Setpoints: IBRs are commonly interfaced with renewable energy sources such as solar and wind. Due to varying environmental conditions, e.g., solar irradiance and wind speed, IBRs exhibit intermittent behavior. As these factors primarily influence the active power output of IBRs, variations in the active power reference P_{ref} need to be considered. The pole loci for different P_{ref} values (from 0 to 1 p.u.) under both weak and strong grid conditions are shown in Fig. 10 and Fig. 11. The default parameters are listed in TABLE I with D_p= 0.04 p.u. and k_{pdc}= 0.1 p.u., and equilibrium points are computed using the equations in *1)*.

Fig. 10 shows the system pole loci with varying P_{ref} under weak grid conditions. As power setpoints increase, the SSO mode conjugate poles $\lambda_{1\text{-}2}$ move towards the RHP but remain within the left-half plane (LHP), with a decrease in oscillation frequency. The similar trends in Fig. 10(a), (b), and (c) illustrate the minimal impact of SDMs on system stability in a weak grid. In contrast, Fig. 11 highlights significant differences under stiff grid conditions. The GFM IBR without SDMs demonstrates strong stability robustness against variations in power setpoints, with a gain margin around 4 dB and a phase margin around 80°. However, the DSOGI- and NF-based GFM IBRs with DVC fail to maintain equilibrium points in these conditions, with $\lambda_{1\text{-}2}$ all located in the RHP. Besides, minor shifts in pole trajectories are observed with variations in power setpoints in stiff grids.

4) Brief Discussion: Despite small-signal stability of GFM IBRs with DVC is slightly influenced by SDMs in weak grids, a stiff grid condition considerably undermines system stability. Referring to the proposed model in Fig. 2, the primary adverse impacts of SDMs on system stability arise from the matrix $\boldsymbol{H}$. This matrix alters the frequency response of the overall control loops, including DVC, APC, RPC, VVC, and VCC. Compared to the DSOGI-based GFM IBR with DVC, the NF-based one shows a better stability boundary. As elucidated in (20) and (25), the system dynamics undergo remarkable restructuring with the high-order and coupled $\boldsymbol{H}^{DSOGI}$, while are only marginal reshaped by $\boldsymbol{H}^{NF}$. While selecting a moderate k_{pdc} value allows the NF-based GFM IBR with DVC to avoid SSO issues under strong grid conditions, achieving robust stability with a high k_{pdc} value remains challenging. Notably, a large k_{pdc} is generally preferred to enhance transient stability and prevent loss of synchronism in GFM IBRs with DVC [20]. Therefore, additional measures are necessary to ensure stable operation of sequence-decomposed GFM IBRs with DVC.

IV. Damping Enhancement Control Method for Sequence-Decomposed GFM IBRs with DVC

This section explores potential solutions to mitigate SSO issues inherent in sequence-decomposed GFM IBRs with DVC under strong grid conditions.

1) Basic Principle: Motivated by the robust stability of GFL IBRs in stiff grids, unifying GFM and GFL controls shows a promising prospect for bolstering system stability amidst variations in AC grid strength. In the typical control architecture of GFL IBRs, the *d*-axis current reference of VCC is generally derived from the active power reference P_{ref} [22], [28]. Inspired by this concept, the active power reference generated by DVC can be introduced as a feedforward loop to $\boldsymbol{I}_{Ldref}^{pc}$ with a positive proportional gain k_{pref}, as shown in Fig. 12. To mitigate steady-state impacts of this feedforward loop, the error signal P_{ref}–P_s is fed forward to $\boldsymbol{I}_{Ldref}^{pc}$ via a high-pass filter $s/(s+\omega_h)$, where ω_h is exemplified as 0.5 Hz in this study. This modification allows GFM IBRs to partially emulate the behavior of GFL IBRs, potentially ensuring stable operation under strong grid conditions.

2) Damping Enhancement Mechanism: Based on (1) and (5), the mechanism underlying the proposed method to enhance stability robustness can be well elucidated. Assuming that a DC-link voltage disturbance $\tilde{V}_{dc}$> 0 occurs, the resultant P_{dc}

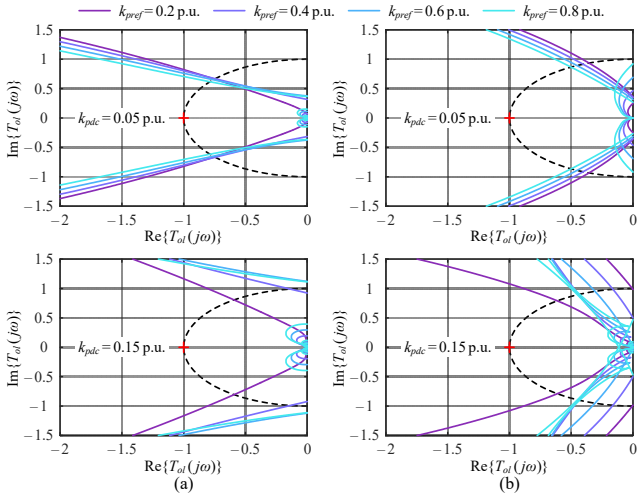


Fig. 14. Nyquist plots for the DSOGI-based GFM IBR with the proposed P_{ref} feedforward control (V_{dcref}= 2.578 p.u., P_{ref}= 0.7 p.u., Q_{ref}= 0). (a) Weak grid condition, L_g= 0.85 p.u. (b) Strong grid condition, L_g= 0.2 p.u.

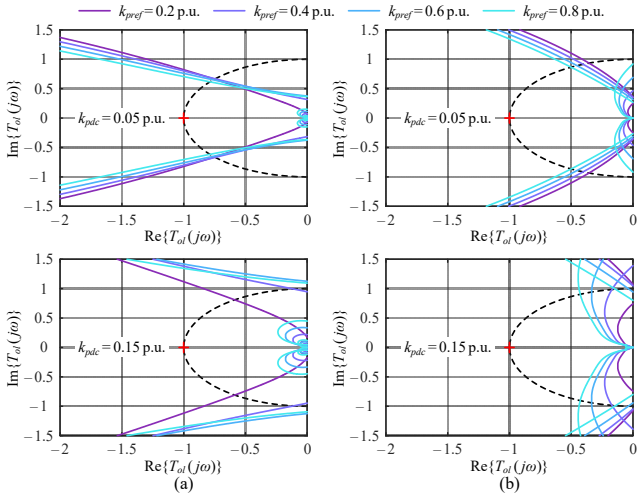


Fig. 15. Nyquist plots for the NF-based GFM IBR with the proposed P_{ref} feedforward control (V_{dcref}= 2.578 p.u., P_{ref}= 0.7 p.u., Q_{ref}= 0). (a) Weak grid condition, L_g= 0.85 p.u. (b) Strong grid condition, L_g= 0.2 p.u.

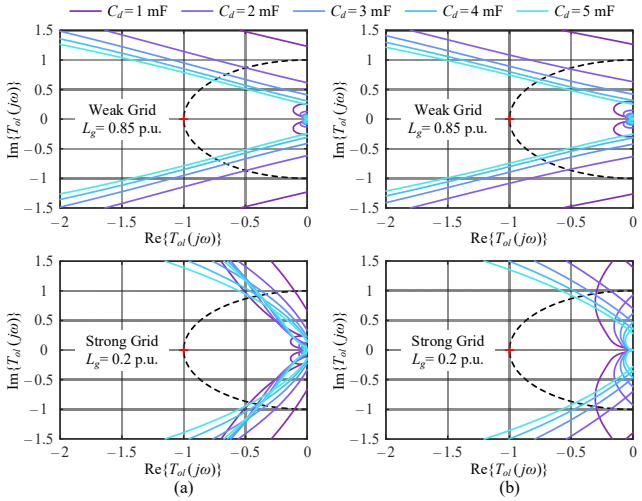


Fig. 16. Nyquist plots for the sequence-decomposed GFM IBR with the proposed P_{ref} feedforward control (V_{dcref}= 2.578 p.u., P_{ref}= 0.7 p.u., Q_{ref}= 0, D_p= 0.04 p.u., k_{pdc}= 0.1 p.u., k_{pref}= 0.4 p.u.). (a) DSOGI-based GFM IBR with DVC. (b) NF-based GFM IBR with DVC.

decreases because of $\tilde{P}_{dc}<0$ [see (5)]. As widely defined in the specifications [5]-[7], the internal voltage phasor of the GFM IBR remains constant or nearly constant over a short duration. Therefore, $\tilde{P}_{dc}$ can be approximated as

$$\tilde{P}_{dc} \approx 1.5 V_{d0}^{pc} \tilde{I}_{Ld}^{s} < 0 \tag{49}$$

Concurrently, an increasing V_{dc} gives rise to a larger P_{ref}, i.e., $\tilde{P}_{ref}>0$ [see (1)]. Through the proposed P_{ref} feedforward control, $\tilde{I}_{Ldref}= k_{pref}\tilde{P}_{ref}>0$. Hence, $\tilde{I}_{Ldref}$ counteracts the variation in $\tilde{I}_{Ld}$ from (49), fostering the system damping capability. Likewise, the proposed method can also effectively mitigate the effect of active power disturbances.

Considering the dynamics of P_{ref}, the SISO model developed in Fig. 5 can be restructured to describe the input-output relationship from $(\tilde{V}_{dcref})^2$ to $\tilde{P}_{ref}$. Since the relationships from $(\tilde{V}_{dcref})^2$ to $(\tilde{V}_{dc})^2$ and from $(\tilde{V}_{dcref})^2$ to $\tilde{P}_{ref}$ share the same open-loop transfer function (TF_{ol}) as given in (43), they exhibit identical closed-loop system poles and thus similar damping properties. Consequently, oscillations in $\tilde{P}_{ref}$ are reduced during transients when the proposed control is activated. Additionally, due to the inclusion of the error signal and high-pass filter, the proposed control does not exhibit steady-state effects on P_{ref}.

3) Control Design: By referring to the small-signal dynamics of DVC and APC in Fig. 2, the small-signal expression of P_{ref} is yielded as

$$\tilde{P}_{ref} = \left[s\tilde{\theta} / \left(\omega_b D_p \right) + \tilde{P} \right] \tag{50}$$

By substituting (29) into (50), $\tilde{P}_{ref}$ can be written as

$$\begin{aligned} \tilde{P}_{ref} &= \boldsymbol{G}_{I_g - P_{ref}} \tilde{\boldsymbol{I}}_{gdq}^{s} + G_{\theta - P_{ref}} \tilde{\theta} \\ &\begin{cases} \boldsymbol{G}_{I_g - P_{ref}} = \boldsymbol{G}_{I_g - P_{ac}} \\ G_{\theta - P_{ref}} = s / \left(\omega_b D_p \right) + G_{\theta - P_{ac}} \end{cases} \end{aligned} \tag{51}$$

With (51), the small-signal model of the proposed P_{ref} feedforward control is shown in Fig. 13. By integrating (51) into (36) and following the procedures outlined in Section III.D, the proposed control can be merged into the developed SISO model. This facilitates the parameter design for the proposed control using the open-loop transfer function TF_{ol} in (43).

Fig. 14 and Fig. 15 show the Nyquist plots for TF_{ol} with the proposed control enabled. Both the DSOGI- and NF-based GFM IBRs are analyzed under weak and strong conditions, with k_{pref} ranging from 0.2 p.u. to 0.8 p.u., and k_{pdc} values of 0.05 p.u., 0.15 p.u., respectively. Other default parameters are consistent with those in TABLE I.

In Fig. 14(a), adequate stability margins are seen for both small and large k_{pdc} values in a weak grid. This robustness results from the fact that only the reference for the inner VCC loop requires modification with the proposed method, while the synchronization loop remains intact. As a result, the GFM nature still dominates the system stability in weak grids. The observations from Fig. 14(b) show that the system can easily achieve stable operation (e.g., phase margins exceeding 50°) in a stiff grid with the proposed control, even with a large k_{pdc} value of 0.15 p.u. This mitigates the SSO instability risks for the DSOGI-based GFM IBR with DVC discussed in Section

III.E. In comparison, the NF-based GFM IBR shows similar Nyquist curves under weak grid conditions, as presented in Fig. 15(a), indicating comparable stability margins to those in Fig. 14(a). However, the open-loop transfer function for the NF-based GFM IBR provides better stability margins under strong grid conditions, particularly with a high k_{pdc} value, achieving a minimal gain margin of 28 dB and phase margin of 64° (with k_{pref}= 0.8 p.u.).

The absence of RHP closed-loop poles in all cases suggests a wide range for k_{pref}, from 0.2 p.u. to 0.8 p.u. To aviod excessive *d*-axis current references during transients and ensure sufficient stability margins, a k_{pref} value of 0.4 p.u. is recommended based on Fig. 14 and Fig. 15.

As indicated in (5), the DC-link dynamics are influenced by the DC capacitor size C_d. Fig. 16 shows the Nyquist plots for TF_{ol} in (43) with k_{pref} = 0.4 p.u. for various DC capacitor sizes. A range of C_d values from 1 mF to 5 mF are considered. None of the Nyquist curves imply RHP poles for the closed-loop system, and adequate stability margins are seen for all cases.

To illustrate the impedance shaping effect of the proposed P_{ref} feedforward control, Bode plots are applied to $\boldsymbol{Z}_{out}$ in (44) to obtain the corresponding impedance profiles. Given that Y_{dd}^{pn}= Y_{dq}^{pn}= Y_{qd}^{pn}= Y_{qq}^{pn}= 0 and Y_{dd}^{np}= Y_{dq}^{np}= Y_{qd}^{np}= Y_{qq}^{np}= 0 in (44), only the impedances Z_{dd}^{pp}, Z_{dq}^{pp}, Z_{qd}^{pp}, Z_{qq}^{pp}, Z_{dd}^{nn}, Z_{dq}^{nn}, Z_{qd}^{nn}, and Z_{qq}^{nn} need to be computed.

Fig. 17 shows the impedance profiles of the DSOGI-based GFM IBR with DVC. The imaginary part of Z_{dd}^{pp} is inductive in the low-frequency range when k_{pref}= 0 but becomes capacitive as k_{pref} increases from 0.1 p.u. to 0.8 p.u. The proposed control significantly reduces the frequency range of negative resistance in Z_{dq}^{pp}, Z_{qd}^{pp}, and Z_{qq}^{pp}, indicating its ability to mitigate instability risks by reshaping the output impedance, as negative resistance can induce unstable oscillations in GFM IBRs with DVC [16], [19]. Notably, the negative sequence impedances Z_{dd}^{nn}, Z_{dq}^{nn}, Z_{qd}^{nn}, and Z_{qq}^{nn} in Fig. 17 remain unchanged despite variations in k_{pref}. This is because the power calculation module for APC and RPC involves only positive sequence components, rendering the outer loops (DVC, APC, and RPC) negligible in their impact on negative sequence impedances. As a result, negative sequence impedances are primarily influenced by the control parameters of VVC and VCC. Since the proposed control alters only the input references for VCC, it does not affect the negative sequence impedance profiles.

The impedance-shaping effects seen in Fig. 17 also apply to the NF-based GFM IBR shown in Fig. 18, which displays similar impedance profiles. However, Fig. 18 shows a reduced frequency range of negative resistance in Z_{dq}^{pp}, Z_{qd}^{pp}, and Z_{qq}^{pp} compared to Fig. 17, even without the proposed control (i.e., with k_{pref}= 0), implying lower instability risks for the NF-based GFM IBR with DVC. This observation aligns with the theoretical analysis in Section III.E, where the DSOGI-based GFM IBR with DVC is noted to be more prone to losing equilibrium points under strong grid conditions, whereas the NF-based GFM IBR with DVC can operate stably in a stiff grid with moderate control parameters.

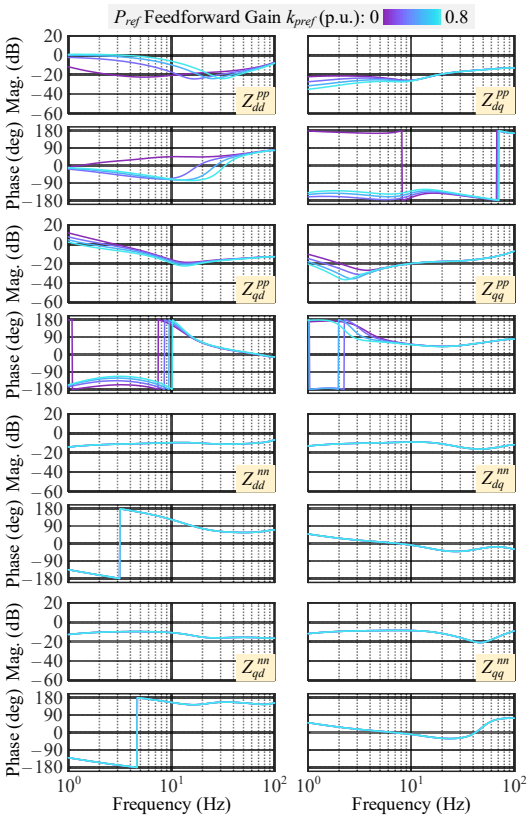


Fig. 17. Impedance profiles for the DSOGI-based GFM IBR with varying k_{pref} (V_{dcref}= 2.578 p.u., P_{ref}= 0.7 p.u., Q_{ref}= 0, D_p= 0.04 p.u., k_{pdc}= 0.1 p.u.).

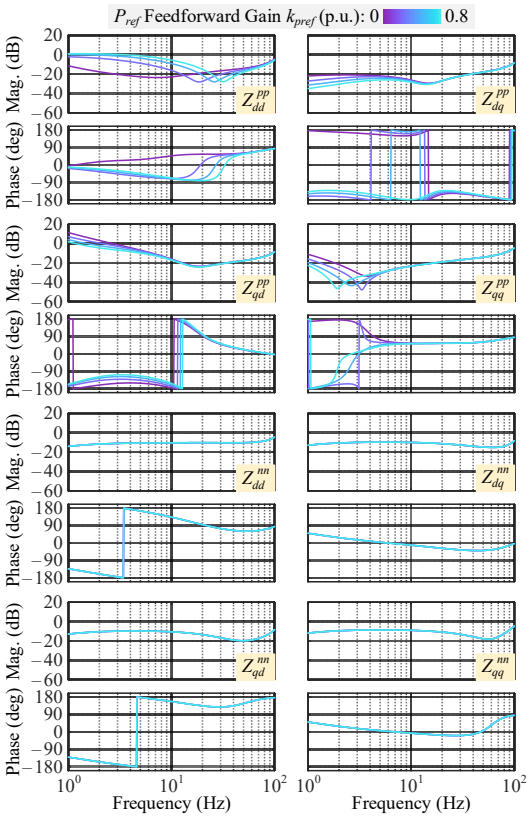


Fig. 18. Impedance profiles for the NF-based GFM IBR with varying k_{pref} (V_{dcref}= 2.578 p.u., P_{ref}= 0.7 p.u., Q_{ref}= 0, D_p= 0.04 p.u., k_{pdc}= 0.1 p.u.).

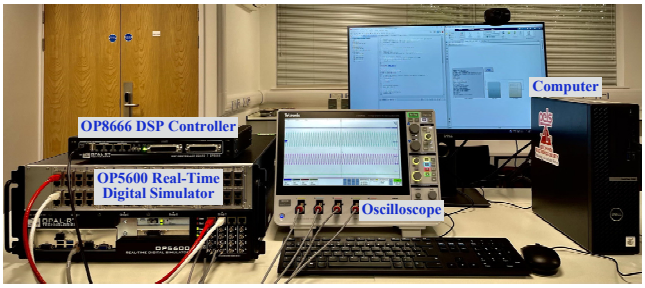


Fig. 19. HIL test setup based on the OPAL-RT platform.

V. Experimental Verification

As shown in Fig. 19, the CHIL test setup is based on OPAL-RT facilities and used for evaluation of the system stability comprising the sequence-decomposed GFM IBR with DVC. The grid-connected system in Fig. 1 is emulated in OP5600, and control algorithms are implemented in TMS320F28379D-based OP8666. Analog signals obtained from OP5600 are fed into OP8666, which then generates PWM digital signals back to OP5600. Parameters outlined in TABLE I are opted for the setup, where both DSOGI- and NF-based GFM IBRs with DVC are considered for illustration.

Fig. 20 shows the step response of the DC-link voltage, comparing experimental results with those calculated from the closed-loop transfer function TF_{cl} in (43). The close alignment between the experimental and calculated curves confirms the accuracy of the SISO model developed in Section III. Thus, the proposed SISO analysis method can effectively predict the stability performance of the closed-loop system.

As discussed in Section III.E, sequence-decomposed GFM IBRs with DVC are prone to significant SSO issues under strong grid conditions. To illustrate these instability risks, different control gains for the DVC and APC, along with diverse power setpoints are considered in the experimental test.

Fig. 21 displays the experimental results showcasing the impact of the DVC proportional gain (k_{pdc}). With the proposed P_{ref} feedforward control activated, the system remains stable across both small and large k_{pdc} values, validating the effectiveness of the proposed control in resisting varying k_{pdc} values. However, when the proposed control is disabled (i.e., k_{pref}= 0), unstable oscillations at 7.6 Hz occur even with a low k_{pdc} value of 0.05 p.u., as shown in Fig. 21(a). Additionally, low-frequency divergent oscillations at 9.6 Hz arise in Fig. 21(b) due to the high DVC gain. These observations align with the theoretical results provided in Fig. 7, indicating that in a stiff grid, equilibrium points do not exist for the DSOGI-based GFM IBR, while the NF-based GFM IBR struggles to operate with a large k_{pdc} value.

The impact of APC droop gains is illustrated in Fig. 22. Both DSOGI- and NF-based GFM IBR operate stably with k_{pref}= 0.4 p.u, while disabling the proposed control leads to increasing oscillations. The observed oscillations in Fig. 22(a) and Fig. 22(b) correspond to the predictions in Fig. 9(b) and Fig. 9(c), respectively, further verifying the accuracy of the developed SISO analysis method.

Fig. 23 demonstrates system dynamics under varying active power setpoints. Successive step changes in P_{ref} are applied to

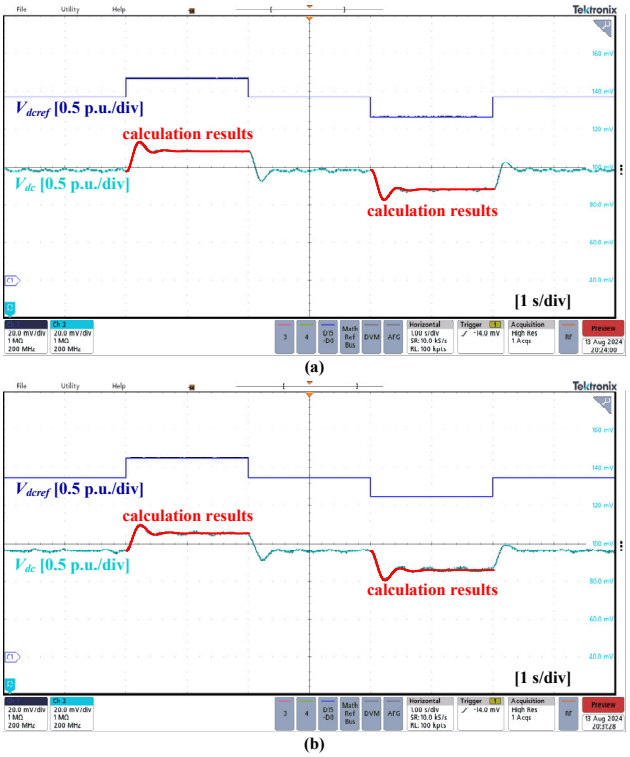


Fig. 20. Experimental results considering step changes of the DC-link voltage reference under the weak grid condition (P_{ref}= 0.7 p.u., Q_{ref}= 0, L_g= 0.85 p.u., D_p= 0.04 p.u., k_{pdc}= 0.1 p.u.). (a) DSOGI-based GFM IBR with DVC. (b) NF-based GFM IBR with DVC.

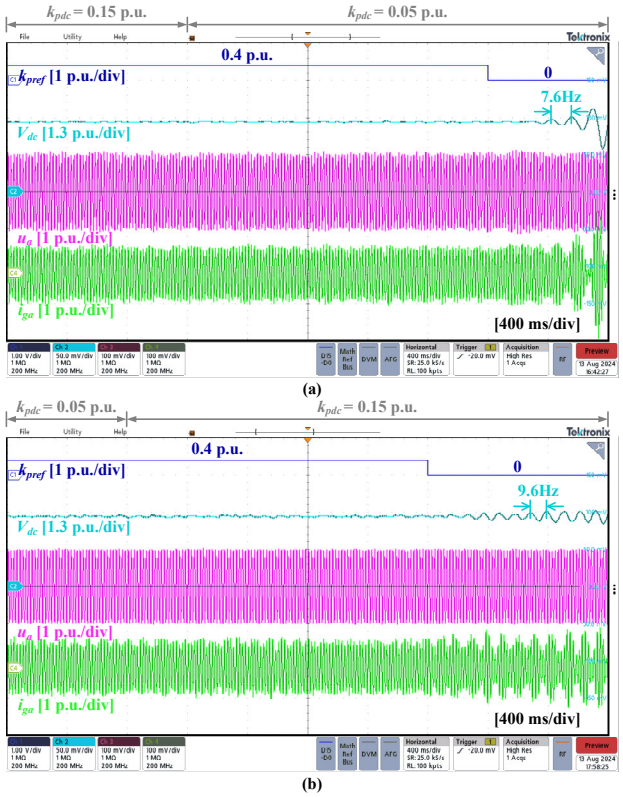


Fig. 21. Experimental results considering impacts of the DVC proportional gains under the strong grid condition (V_{dcref}= 2.578 p.u., P_{ref}= 0.7 p.u., Q_{ref}= 0, L_g= 0.2 p.u., D_p= 0.04 p.u.). (a) DSOGI-based GFM IBR with DVC. (b) NF-based GFM IBR with DVC.

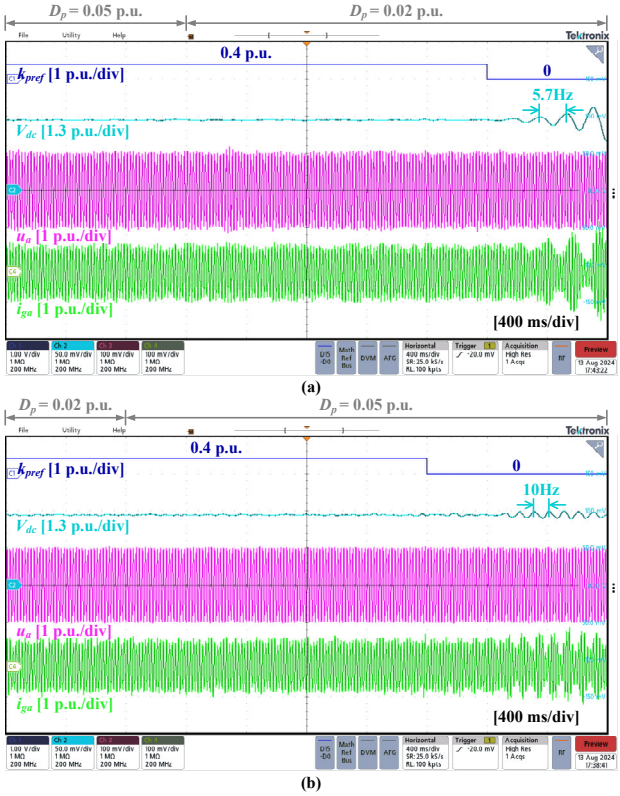


Fig. 22. Experimental results considering impacts of the APC droop gains under the strong grid condition (V_{dcref}= 2.578 p.u., P_{ref}= 0.7 p.u., Q_{ref}= 0, L_g= 0.2 p.u., k_{pdc}= 0.1 p.u.). (a) DSOGI-based GFM IBR with DVC. (b) NF-based GFM IBR with DVC.

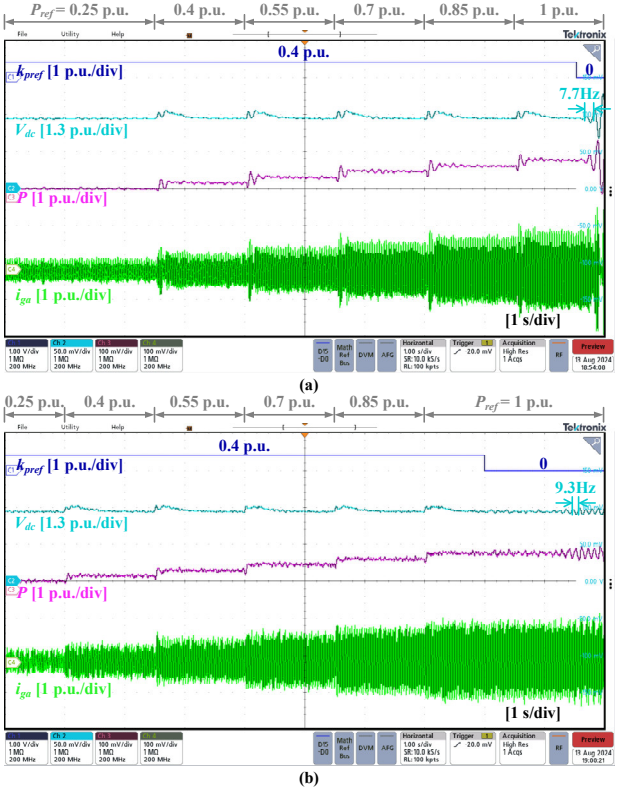


Fig. 23. Experimental results considering impacts of active power setpoints under the strong grid condition (V_{dcref}= 2.578 p.u., Q_{ref}= 0, L_g= 0.2 p.u., D_p= 0.04 p.u., k_{pdc}= 0.1 p.u.). (a) DSOGI-based GFM IBR with DVC. (b) NF-based GFM IBR with DVC.

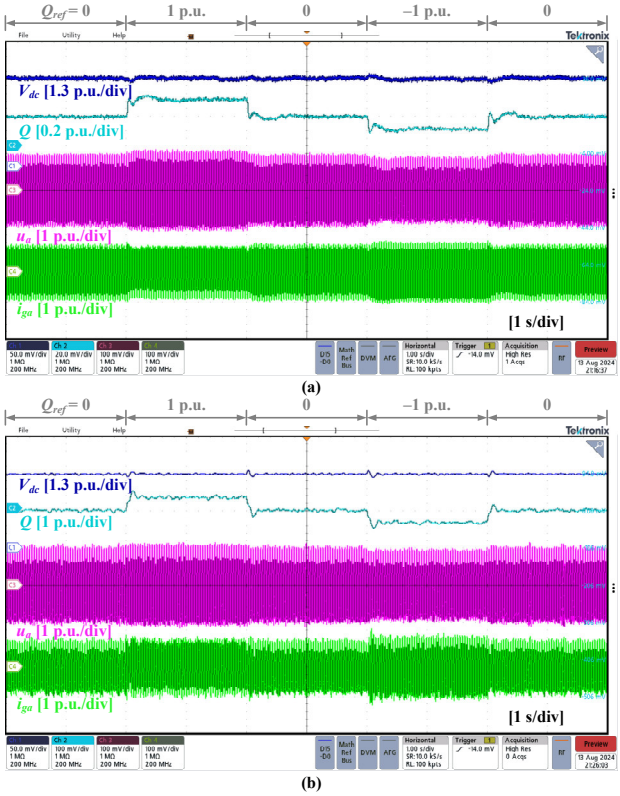


Fig. 24. Experimental results with varying reactive power references (V_{dcref}= 2.578 p.u., P_{ref}= 0.7 p.u., D_p= 0.04 p.u., k_{pdc}= 0.1 p.u.). (a) Weak grid condition, L_g= 0.85 p.u. (b) Strong grid condition, L_g= 0.2 p.u.

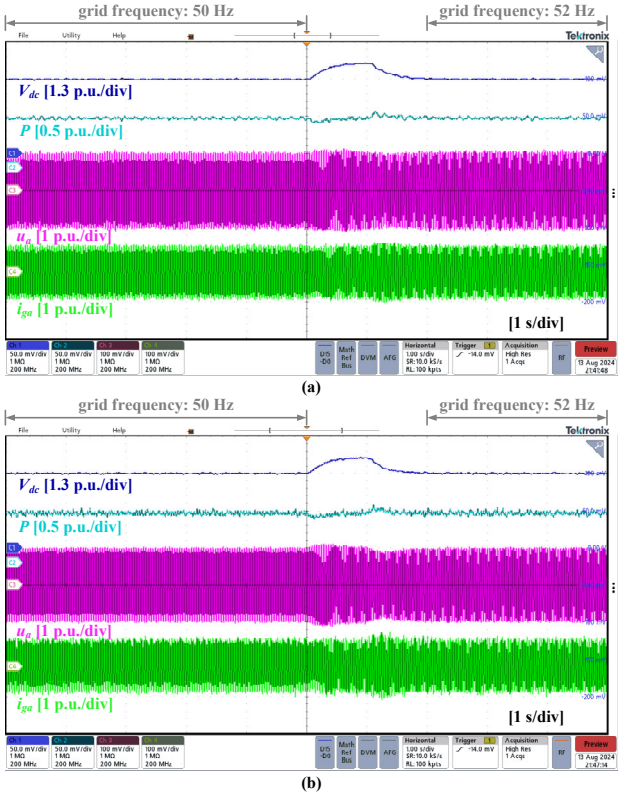


Fig. 25. Experimental results under 2 Hz·s^{-1} grid frequency variation (V_{dcref}= 2.578 p.u., P_{ref}= 0.7 p.u., Q_{ref}= 0, L_g= 0.3 p.u., D_p= 0.04 p.u., k_{pdc}= 0.1 p.u., k_{pref}= 0.4 p.u.). (a) DSOGI-based GFM IBR with DVC. (b) NF-based GFM IBR with DVC.

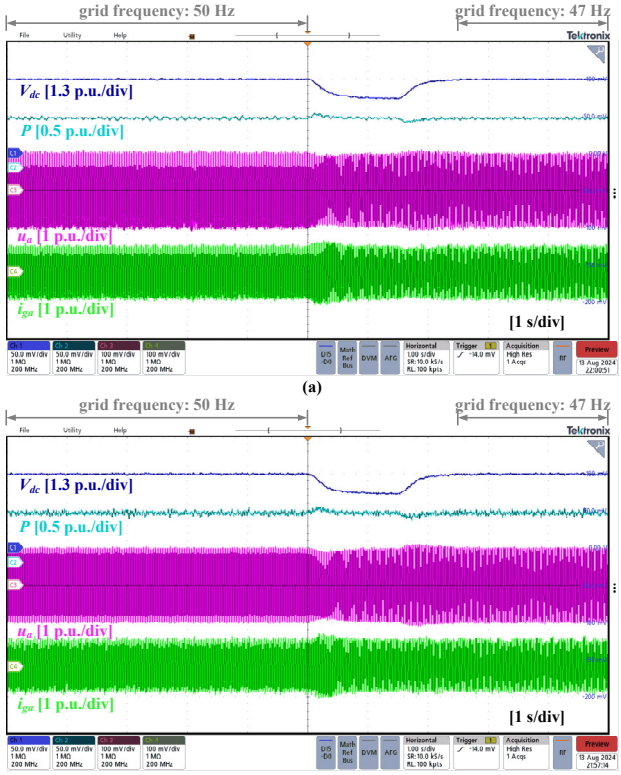


Fig. 26. Experimental results under –2 Hz·s^{-1} grid frequency variation (V_{dcref}= 2.578 p.u., P_{ref}= 0.7 p.u., Q_{ref}= 0, L_g= 0.3 p.u., D_p= 0.04 p.u., k_{pdc}= 0.1 p.u., k_{pref}= 0.4 p.u.). (a) DSOGI-based GFM IBR with DVC. (b) NF-based GFM IBR with DVC.

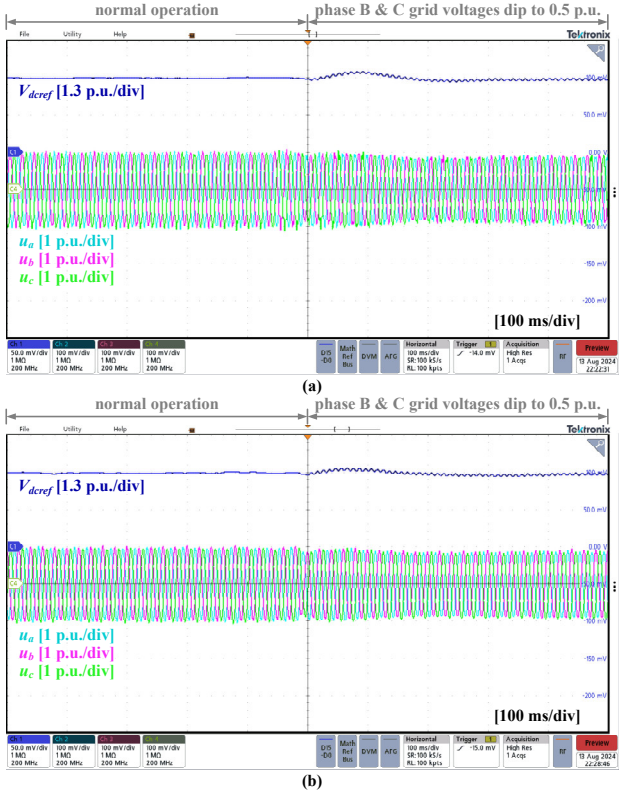


Fig. 27. Experimental results under asymmetric voltage sags (V_{dcref}= 2.578 p.u., P_{ref}= 0.7 p.u., Q_{ref}= 0, L_g= 0.3 p.u., D_p= 0.04 p.u., k_{pdc}= 0.1 p.u., k_{pref}= 0.4 p.u.). (a) DSOGI-based GFM IBR with DVC. (b) NF-based GFM IBR with DVC.

sequence-decomposed GFM IBRs with DVC. With the proposed control enabled, the system is always in stable operation irrespective of the P_{ref} values, yet oscillations appear in both V_{dc} and P waveforms once k_{pref} is set to 0. The SSO frequencies reflected in Fig. 23 closely match the theoretical results presented in Fig. 11.

To evaluate the performance of the proposed control under varying reactive power responses, a range of Q_{ref} values from –1 p.u. to 1 p.u. is applied in Fig. 24, using a DSOGI-based GFM IBR in both weak and stiff grids, with D_p= 0.04 p.u. and k_{pdc}= 0.1 p.u. As shown in Fig. 24, the system remains stable even with significant variations in Q_{ref}, and the Q-V droop law actively responds throughout the process.

Grid-side events, including frequency and voltage disturbances, are also tested. Both DSOGI- and NF-based GFM IBRs with DVC are used in the case study, with D_p= 0.04 p.u., k_{pdc}= 0.1 p.u., k_{pref}= 0.1 p.u., under a moderate SCR of 3.3. Fig. 25 shows system dynamics under a grid frequency change from 50 Hz to 52 Hz (RoCoF: 2Hz·s^{-1}). The instantaneous active power exchange between the sequence-decomposed GFM IBR and the AC grid results in an increased V_{dc}. With DVC and APC regulation, the system eventually stabilizes in a 52 Hz AC system. An inverse trend in V_{dc} and P is observed in Fig. 26. When the grid frequency changes from 50 Hz to 47 Hz, the sequence-decomposed GFM IBR maintains system stability and adapts to the 47 Hz AC grid, confirming the proposed control's feasibility during frequency disturbances.

Finally, asymmetrical voltage sags are applied to further validate the proposed control. A severe two-phase grid voltage dip from 1 p.u. to 0.5 p.u. is induced for phase B and phase C. Parameters from Figs. 25 and 26 are retained. The successful fault ride-through in Fig. 27 demonstrates superior system stability with the proposed control active under asymmetric sag conditions. Additionally, the PCC voltage exhibits balanced waveforms with lifted magnitude, consistent with the sequence-decomposed VVC control settings in (34). Here, only the d-axis positive sequence voltage follows the voltage reference from the RPC, while U_q^{pc}, U_d^{nc}, and U_q^{pc} are set to 0. The observed second-order harmonics in V_{dc} stem from unbalanced active power transfer. These results verify the effectiveness of the proposed control under asymmetrical voltage events.

VI. Conclusion

The deployment of the sequence-decomposed control in GFM IBRs inevitably introduces SDMs into the GFM control system, thereby altering the system dynamics. To investigate the resultant changes in system stability, this paper develops a generalized SISO small-signal model for the sequence-decomposed GFM IBR with DVC. Compared to GFM IBRs without SDMs, those with SDMs show nearly identical or slightly improved stability within the SSO mode in a weak grid. Whereas sequence-decomposed GFM IBRs with DVC tend to lose their steady-state operating points in a stiff grid and are prone to SSO instability issues, especially when configured with the DSOGI-based SDM. Inspired by the robust stability of

GFL IBRs under strong grid conditions, an active power reference feedforward method is proposed to enhance system damping capability. It has been demonstrated via extensive CHIL tests that the reported method effectively contributes to the stable operation of sequence-decomposed GFM IBRs with DVC under various operating conditions and grid events. In some cases, frequency coupling effects induced by SDMs are significant and may interact with those caused by asymmetric outer control loops. This complex mechanism necessitates further investigation in future research.

REFERENCES

[1] J. MacDowell *et al.*, "Grid-forming technology in energy systems integration," Energy Systems Integration Group, Tech. Rep., Mar. 2022.

[2] A Johnson, "Minimum specification required for provision of GB grid forming (GBGF) capability (formerly virtual synchronous machine/VSM capability)," National Grid ESO, Final Modification Rep. GC0137, Nov. 2021.

[3] Y. Lin *et al.*, "Research roadmap on grid-forming inverters," National Renewable Energy Laboratory, Golden, REL/TP-5D00-73476, 2020.

[4] "Great Britain grid forming best practice guide," National Grid ESO, Apr. 2023.

[5] "Voluntary specification for grid-forming inverters," Australian Energy Market Operator, May 2023.

[6] B. Kroposki *et al.*, "Specifications for grid-forming inverter-based resources-Version 1," UNIFI Technical Report, UNIFI-2022-2-1, Dec. 2022.

[7] "White paper: Grid forming functional specifications for BPS-connected battery energy storage systems," North American Electric Reliability Corporation, Sep. 2023.

[8] H. Antti, "Specific study requirements for grid energy storage systems," FINGRID, Jun. 2023.

[9] M.-A. Nasr and A. Hooshyar, "Controlling grid-forming inverters to meet the negative-sequence current requirements of the IEEE Standard 2800-2022," *IEEE Trans. Power Deliv.*, vol. 38, no. 4, pp. 2541–2555, Aug. 2023.

[10] D. Ramasubramanian *et al.*, "Preliminary gap analysis of existing IEEE 1547 and IEEE 2800 Standards towards GFM technology," National Renewable Energy Laboratory, Golden, NREL/TP-5D00-88609, Feb. 2024.

[11] Z. Li, K. W. Chan, J. Hu, and S. W. Or, "An adaptive fault ride-through scheme for grid-forming inverters under asymmetrical grid faults," *IEEE Trans. Ind. Electron.*, vol. 69, no. 12, pp. 12912–12923, Dec. 2022.

[12] M. A. Awal, M. R. K. Rachi, H. Yu, I. Husain, and S. Lukic, "Double synchronous unified virtual oscillator control for asymmetrical fault ride-through in grid-forming voltage source converters," *IEEE Trans. Power Electron.*, vol. 38, no. 6, pp. 6759–6763, Jun. 2023.

[13] H. Zhang, R. Liu, C. Xue, and Y. Li, "Simultaneous overvoltage and overcurrent mitigation strategy of grid-forming inverters under a single-line-to-ground fault," *IEEE Trans. Ind. Electron.*, vol. 71, no. 9, pp. 10818–10830, Sep. 2024.

[14] M. A. Awal, S. Cen, M. R. K. Rachi, H. Yu, S. Schröder, and I. Husain, "Modeling, analysis, and design for small-signal stability in sequence-decomposed grid-forming control," *IEEE Trans. Ind. Appl.*, vol. 60, no. 1, pp. 865–875, Jan. 2024.

[15] Z. Zhao, P. Yang, Y. Wang, Z. Xu, and J. M. Guerrero, "Dynamic characteristics analysis and stabilization of PV-based multiple microgrid clusters," *IEEE Trans. Smart Grid*, vol. 10, no. 1, pp. 805–818, Jan. 2019.

[16] L. Zhao, X. Wang, and Z. Jin, "Impedance-based dynamics analysis for DC-link voltage-synchronized voltage-source converters," *IEEE Trans. Power Electron.*, vol. 38, no. 9, pp. 10829–10844, Sep. 2023.

[17] L. Zhao, Z. Jin, and X. Wang, "Small-signal synchronization stability of grid-forming converters with regulated DC-link dynamics," *IEEE Trans. Ind. Electron.*, vol. 70, no. 12, pp. 12399–12409, Dec. 2023.

[18] L. Zhao, Z. Jin, and X. Wang, "Analysis and damping of low-frequency oscillation for DC-link voltage-synchronized VSCs," *IEEE Trans. Power Electron.*, vol. 38, no. 7, pp. 8177–8189, Jul. 2023.

[19] J. Guo *et al.*, "Impedance analysis and stabilization of virtual synchronous generators with different dc-link voltage controllers under weak grid," *IEEE Trans. Power Electron.*, vol. 36, no. 10, pp. 11397–11408, Oct. 2021.

[20] C. Luo, T. Liu, X. Wang, and X. Ma, "Design-oriented analysis of DC-link voltage control for transient stability of grid-forming inverters," *IEEE Trans. Ind. Electron.*, vol. 71, no. 4, pp. 3698–3707, Apr. 2024.

[21] M. A. Awal, M. R. K. Rachi, H. Yu, S. Schroder, and I. Husain, "Symmetrical components extraction for grid-forming voltage source converters," in *2022 IEEE Energy Conversion Congress and Exposition (ECCE)*, Detroit, MI, USA: IEEE, Oct. 2022, pp. 1–8.

[22] Q. Liu, K. Jia, B. Yang, L. Zheng, and T. Bi, "Fault analysis of inverter-interfaced RESs considering decoupled sequence control," *IEEE Trans. Ind. Electron.*, vol. 70, no. 5, pp. 4820–4830, May 2023.

[23] S. Golestan, M. Monfared, and F. D. Freijedo, "Design-oriented study of advanced synchronous reference frame phase-locked loops," *IEEE Trans. Power Electron.*, vol. 28, no. 2, pp. 765–778, Feb. 2013.

[24] M. Zubiaga *et al.*, "D3.3: Analysis of the synchronisation capabilities of BESS power converters," OSMOSE Project, Mar. 2022.

[25] H. Zhang, L. Harnefors, X. Wang, H. Gong, and J.-P. Hasler, "Stability analysis of grid-connected voltage-source converters using SISO modeling," *IEEE Trans. Power Electron.*, vol. 34, no. 8, pp. 8104–8117, Aug. 2019.

[26] F. Cecati and M. Liserre, "Stability analysis of converter-dominated grid using nonlinear grey-box converter models," May 22, 2024. doi: 10.36227/techrxiv.171639117.73358192/v1.

[27] *IEEE standard for interconnection and interoperability of inverter-based resources (IBRs) Interconnecting with associated transmission electric power systems*, IEEE Std 2800-2022, Apr. 2022.

[28] L. Harnefors, M. Bongiorno, and S. Lundberg, "Input-admittance calculation and shaping for controlled voltage-source converters," *IEEE Trans. Ind. Electron.*, vol. 54, no. 6, pp. 3323–3334, Dec. 2007.

[29] L. Zhang, L. Harnefors, and H.-P. Nee, "Power-synchronization control of grid-connected voltage-source converters," *IEEE Trans. Power Syst.*, vol. 25, no. 2, pp. 809–820, May 2010.

[30] P. Rodríguez, A. Luna, R. S. Muñoz-Aguilar, I. Etxeberria-Otadui, R. Teodorescu, and F. Blaabjerg, "A stationary reference frame grid synchronization system for three-phase grid-connected power converters under adverse grid conditions," *IEEE Trans. Power Electron.*, vol. 27, no. 1, pp. 99–112, Jan. 2012.

[31] J. R. Carson, "Notes on the theory of modulation," *Proc. IRE*, vol. 10, no. 1, pp. 57–64, Feb. 1922.

[32] L. Huang, C. Wu, D. Zhou, and F. Blaabjerg, "A simplified SISO small-signal model for analyzing instability mechanism of grid-forming inverter under stronger grid," in *2021 IEEE 22nd Workshop on Control and Modelling of Power Electronics (COMPEL)*, Cartagena, Colombia: IEEE, Nov. 2021, pp. 1–6.

[33] F. Zhao, X. Wang, Z. Zhou, Y. Sun, L. Harnefors, and T. Zhu, "Robust grid-forming control with active susceptance," *IEEE Trans. Power Electron.*, vol. 38, no. 3, pp. 2872–2877, Mar. 2023.

[34] J. Yu, S. Wang, Z. Liu, J. Li, J. Liu, and J. Shang, "Accurate small-signal terminal characteristic model and SISO stability analysis approach for parallel grid-forming inverters in islanded microgrids," *IEEE Trans. Power Electron.*, vol. 38, no. 5, pp. 6597–6612, May 2023.

[35] H. Zhang, X. Wang, L. Harnefors, H. Gong, J.-P. Hasler, and H.-P. Nee, "SISO transfer functions for stability analysis of grid-connected voltage-source converters," *IEEE Trans. Ind. Appl.*, vol. 55, no. 3, pp. 2931–2941, May 2019.

[36] X. Wang, L. Harnefors, and F. Blaabjerg, "Unified impedance model of grid-connected voltage-source converters," *IEEE Trans. Power Electron.*, vol. 33, no. 2, pp. 1775–1787, Feb. 2018.

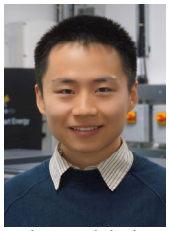
[37] F. Cecati, J. K. M. Becker, S. Pugliese, Y. Zuo, M. Liserre, and M. Paolone, "LTP modeling and analysis of frequency coupling in PLL-synchronized converters for harmonic power flow studies," *IEEE Trans. Smart Grid*, vol. 14, no. 4, pp. 2890–2902, Jul. 2023.

[38] X. Wang and F. Blaabjerg, "Harmonic stability in power electronic-based power systems: Concept, modeling, and analysis," *IEEE Trans. Smart Grid*, vol. 10, no. 3, pp. 2858–2870, May 2019.

[39] L. Beloqui Larumbe, Z. Qin, L. Wang, and P. Bauer, "Impedance modelling for three-phase inverters with double synchronous reference frame current controller in the presence of imbalance," *IEEE Trans. Power Electron.*, pp. 1461–1475, Feb. 2022.

[40] L. Harnefors, M. Hinkkanen, U. Riaz, F. M. M. Rahman, and L. Zhang, "Robust analytic design of power-synchronization control," *IEEE Trans. Ind. Electron.*, vol. 66, no. 8, pp. 5810–5819, Aug. 2019.

[41] D. G. Holmes, T. A. Lipo, B. P. McGrath, and W. Y. Kong, "Optimized design of stationary frame three phase AC current regulators," *IEEE Trans. Power Electron.*, vol. 24, no. 11, pp. 2417–2426, Nov. 2009.



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